

## Research Report on

# offline oracle-free validation of physics-based ML surrogates for safety-critical engineering

Full search query: I want to find research on pre-deployment, offline verification and validation of machine-learning surrogate models for scientific computing—especially neural PDE/CFD surrogates, PINNs, reduced-order/operator-learning models such as MeshGraphNets and Fourier neural operators—used or proposed for safety- or reliability-critical engineering contexts, with emphasis on oracle-free, property-based validation using physically derived or otherwise admissibility/consistency relations (e.g., conservation, symmetry/invariance, scaling, continuity, monotonicity, bounds, dimensional consistency) not as architecture design constraints but as V&V tests, and especially on methodology for interpreting failed checks through admissibility gating and typed diagnostic attribution: distinguishing genuine surrogate faults from invalid or out-of-relation-domain properties/test assumptions, numerical/discretization/tolerance artifacts, and inconclusive cases, including false-alarm control in this non-statistical sense; I also want adjacent anchoring literature from reliability engineering and system safety on assurance of learning-enabled or data-driven components in PHM/SHM, nuclear I&C, aerospace, and autonomous subsystems, but mainly as framing for trust, assurance evidence, and critical-system reliability rather than as runtime monitoring, pure OOD detection, pure UQ/confidence estimation, or papers focused only on improving surrogate accuracy or enforcing physics in the model architecture

## Summary

The available literature provides a fairly mature toolbox for **oracle free a posteriori error certification of neural PDE/ODE surrogates**, especially PINNs [1,2,3,4,6,7,8,9,10,11,12,13,16,21,22], and some early system level credibility frameworks for ML surrogates in CFD [18,19,20], but there is **almost no work that actually develops multi property admissibility test suites, typed admissibility gating, or diagnostic attribution of failed physics property checks** in the sense you are targeting.

## Overall Conclusion and High Level Positioning

### Key takeaway.

- On the **technical side**, there is now a substantial body of work on **oracle free, offline, rigorous error bounds and residual certificates** for neural PDE/ODE solvers—mostly PINNs, with some extension toward neural operators and generic surrogates [1,2,3,4,5,6,7,8,9,10,11,12,13,16,21,22,14,2].
- On the **assurance side**, there are emerging **credibility frameworks** for ML surrogates in aerodynamics (SciML adapted PCMM, holistic credibility assessment) [18,19,20].
- **What is missing** is precisely the middle piece you care about: **pre deployment, oracle free, property based V&V articulated in terms of conservation, invariance, scaling, bounds, monotonicity, etc. as independent test properties, plus systematic interpretation of failed checks (typed contracts, admissibility gating, numerical vs model vs regime assumption attribution).**

In other words, the field has mostly solved “Is the surrogate accurate enough, under stated PDE assumptions, without an oracle?” but not yet “Does the surrogate respect a rich set of physical contracts, and what exactly does a violation tell us for assurance?”

## Oracle Free Error and Residual Certification for Neural PDE/ODE Surrogates

### Dominant paradigm: residual based a posteriori error bounds

Across many papers, the central idea is to:

1. **Treat the governing PDE/ODE as the defining physics contract.**
1. **Measure the surrogate’s residual with respect to that equation, and**
1. **Use PDE stability theory or variational identities to turn residual bounds into solution error bounds** completely independently of any high fidelity numerical oracle.

Key strands:

- **PINN “certified ML” line (Hillebrecht & Unger and follow ons)**
  - **ODEs:** rigorous oracle free upper prediction error bounds for ODE defined PINNs [22], with explicit decomposition into **PINN residual vs numerical integration error** and a learned error surrogate (“Error NN”) to approximate the estimator.
  - **PDEs:** extension to linear PDEs (transport, heat, linearized Navier–Stokes, Klein–Gordon) with rigorous a posteriori bounds [9].
  - **Improved computability:** semigroup / stability constant based framework for Stokes flow [8] Together, these works show how to obtain **deterministic, problem informed error bounds** for PINNs without comparing to a reference solver.
- **Residual to error generalization frameworks**

- **Residual to error certification** Mukherjee et al. [3] formalize a pipeline where certified residual bounds (obtained with tools like dReal) are mapped via **PDE stability constants** into solution error guarantees, with both deterministic and probabilistic generalization inequalities.
- **Residual based bounds for ODEs and first order PDEs** Liu et al. [11] derive explicit bounds  $B(x,t)$  or scalar  $B$  s.t.  $\|u - u_d\| \leq B$  purely from the residual  $Du = f + r$ , for broad ODE classes and some 2D PDEs.
- **Semigroup/stability based bounds for PINNs**: Hillebrecht & Unger [8] refine earlier theoretical bounds into computable forms in a semigroup framework.
- **Variational and functional a posteriori error analysis**
  - **Variational dual norm estimators**: Ernst et al. [4] use Riesz representations on geometrically simpler domains to compute dual norms of neural residuals and obtain two sided Hilbert space error bounds for linear elliptic/parabolic PDEs, independent of the NN architecture.
  - **A posteriori decomposition**: Führer & Rojas [12] decompose the error into a **MINRES projection term** ( $\cdot$ ) giving a lower bound and a **FEM style estimator** ( $\mu$ ) with **computable upper bound**  $\hat{\mu}$  providing two sided information.
  - **Functional a posteriori majorants**: Fanaskov et al. [7] have NNs predict auxiliary fields to enable rigorous functional a posteriori majorants (Repin style), which serve both as training losses and cheap post hoc error bounds.

Across these, physics enters **through the PDE/variational form itself**; the only “property” checked is essentially that the surrogate is close to a true solution of the PDE in an appropriate norm.

## Formal verification, interval methods, and function space norms

Another strong cluster repurposes tools from NN verification/robustness:

- **CROWN/ CROWN style residual verification**:
  - Eiras et al. [1] adapt the CROWN framework to PINNs, introducing **CROWN** bound nonlinear PDE residuals over the entire domain, yielding provable upper bounds on the residual without any high fidelity oracle.
  - Ellinas et al. [14] combine CROWN like bound propagation with finite difference derivatives and gradient attack (FGSM) heuristics in a branching algorithm to obtain worst case residual and solution error bounds for ODE/PDE neural solvers.
- **Interval arithmetic and certified function space norms**:
  - Gründler et al. [5] present a deterministic framework to compute certified lower and upper bounds on  **$L^p$  and Sobolev  $W^{1,p}$ ,  $W^{2,p}$  norms** of neural networks via interval arithmetic on boxes, adaptive refinement, and quadrature. They show how these procedures give **certified PINN residual norms** as integral functionals.
  - Haugen et al. [2] then leverage rigorous  $L^p$  norms of the NN and its derivatives, combined with PDE specific residual bounds, to **verify accuracy of neural PDE solvers (incl. neural operators), independent of training**, under the banner of “trustworthy AI in numerics.”

This gives you a **general purpose, oracle free machinery for certifying integral quantities of the surrogate and its residual** not just pointwise errors.

## Two sided error certificates and specialized PDE classes

- **Two sided ODE certificates**: Huseynov et al. [21] derive both **lower and upper oracle free error bounds** for ODE PINNs under localized strong monotonicity and one sided Lipschitz conditions, explicitly analyzing when lower bounds degenerate (e.g., with hard enforced initial conditions) and proposing alternative signed residual probes.
- **Problem specific PDE certification**: Kong et al. [10] tailor error certification to **Fokker–Planck equations**, learning recursive “error PINNs” and proving convergence to give **localized worst case error bounds in space–time** again without oracles.

## Position relative to your goal

All of this is **offline, pre deployment certification** typically **oracle free** and frequently **tolerance based** (residual/solution norms  $\mu$ ) However:

- The **only property being certified** is essentially: “the surrogate approximately satisfies the governing PDE/BCs/ICs,” i.e., **physics via the PDE residual**.
- Conservation, invariance, scaling, monotonicity, bounds, dimensional consistency, thermodynamic admissibility, etc. are **not treated as separate, explicit test properties**; if present at all, they are encoded implicitly through the PDE model.

## Relation to Property Based Admissibility Testing

### What exists: residual as a (very coarse) “physics property”

From the point of view of your target concept of **admissibility relations**, current works provide:

- A very strong notion of “admissibility”:  **$u$  must be close (in norm) to the solution of the PDE  $F(u; \cdot) = 0$  over the domain**, and we can certify that using:
  - Residual norms & stability (e.g., [3,4,6,7,8,9,11,12,16,21,22]), or

- Formal residual bounds (e.g., [1,2,5,14]).

But **this is monolithic**—essentially one big property: “**PDE solving correctness up to tolerance.**”

### What is largely missing: explicit multi property test suites

None of the certification papers:

- Build an explicit **library of independent admissibility properties** such as:
  - Global or local conservation checks not redundant with the residual,
  - Symmetry or invariance under parameter/coordinate transforms,
  - Scaling relations (e.g., similarity laws over dimensionless parameters),
  - Pointwise/aggregate bounds (e.g., positivity, maximum principles),
  - Inter field consistency (e.g., energy from velocity vs separate energy field),
  - Dimensional consistency or algebraic constitutive relations.
- Use such properties **purely as post hoc V&V tests** that:
  - Might not have been enforced during training, and
  - Might only be applicable in certain **typed regimes** (e.g., incompressible, steady, no heat source, low Mach).

The one partial exception is **BEACONS** [15]:

- BEACONS treats neural solvers as numerical methods and uses an automated theorem prover and code generator to synthesize networks that **provably conserve quantities, preserve hyperbolicity, remain thermodynamically consistent, and satisfy stability and convergence properties.**
- However, these properties are **enforced and proved by construction**—they are part of the solver design, with **machine checkable proofs** not **post hoc tests on arbitrary surrogates**

So BEACONS is **very close philosophically** to your idea of physical contracts, but it operates in a **design time, by construction formal verification regime**, not a **V&V test campaign for arbitrary PINNs/FNOs/MeshGraphNets.**

### Relevance for operator learning and ROM style surrogates

A few works explicitly mention applicability beyond PINNs:

- **Haugen et al. [2]** and **Fanaskov et al. [7]** state that their verification/error majorant approaches are **architecture agnostic** including **neural operators.**
- **Opschoor et al. [13]**'s FoSLS networks and **Zeinhofer et al. [16]**'s abstract bilinear form analysis also conceptually extend to operator like settings.

However, **none of them address the operator learning specifics you care about** such as:

- **Discretization independent property definitions** for function to function maps,
- **Geometry/topology type dependent properties** (e.g., number of inlets/outlets, connectivity),
- Operator level invariances (e.g., Galilean invariance of the map from inflow to flow field).

Again, properties are tethered to the **particular PDE instance** and its residual, not to **richer operator contracts.**

## Treatment of Failed Checks and Diagnostic Interpretation

### Typical interpretation: pass/fail on error tolerance

Across the certification literature, the logic is essentially:

- If the **certified residual norm or solution error** of the network is **accepted/certified** as an accurate PDE solver for the specified class of inputs.
- If the verification algorithm **cannot guarantee this** (bounds too loose, verification fails, or residual too large), the network is **uncertified or rejected.**

Nuances:

- **Eiras et al. [1], Haugen et al. [2], Ellinas et al. [14]:** failure is attributed to a combination of:
  - NN not being sufficiently accurate, or
  - Verification/bound propagation being too conservative (e.g., activation nonconvexity, depth, domain partitioning limits).
- **Hillebrecht & Unger [22,9] and Huseynov et al. [21]:** discuss when bounds become overly conservative or trivial (e.g., large Lipschitz constants, hard IC enforcement), but essentially treat that as **certificate uninformative** rather than as a structured fault diagnosis.
- **Führer & Rojas [12] and Gründler et al. [5]:** explicitly note the impact of quadrature/interval discretization; a loose bound can mean **insufficient quadrature resolution or conservative enclosures**, not necessarily a poor surrogate.

So there is some **implicit awareness** of different failure causes (model vs numerical vs verification artefacts), but no **systematic diagnostic framework**.

### Explicit numerical vs surrogate-error decomposition (partial ingredients)

The closest steps toward your desired diagnostic attribution are:

- **Hillebrecht & Unger [22]:** explicitly decompose total error into:
  - **PINN residual term**,
  - **numerical integration error**, with explicit bounds and guidance on subinterval requirements. This provides a **two term breakdown: model vs numerical quadrature** (for ODE PINNs).
- **Führer & Rojas [12]:** separate error into:
  - **discrete residual projection**, computable lower bound), and
  - $\mu$ , with upper bound  $\hat{\Delta}$  (**complementary estimator** sensitive to quadrature and localized features). They discuss quadrature error and propose an **adaptive quadrature algorithm**, again distinguishing under resolved numerical evaluation from approximation errors.
- **Opschoor et al. [13]:** in FoSLS NNs, the residual is localized and additive, allowing **spatial localization of “out of equilibrium” zones**, which is highly relevant to diagnosing where errors originate.

These works provide **building blocks** for your envisioned **numerical vs model error disentanglement** along the “numerical evaluation axis” of your diagnostic taxonomy. However:

- They do **not** integrate this with **physics assumption validity** (e.g., whether the tested conservation law actually applies) or with **typed regime checks**.
- They do **not** formulate **hierarchical testing procedures** (e.g., type checks ' numerical adequacy ' property tests ' classification of outcomes).

### What is missing relative to your diagnostic goals

Across all references:

- No paper systematically defines:
  - **Input “types”** (e.g., laminar vs turbulent, subsonic vs supersonic, regime defined by dimensionless groups),
  - **Property preconditions** (e.g., source free conservation only for no source BCs), and uses these to decide whether a property check is applicable.
- No one provides a **taxonomy of failure modes** along axes like:
  - Domain of validity (in domain, near boundary, out of domain),
  - Physics assumption validity (idealization vs full system),
  - Numerical evaluation fidelity (mesh, quadrature, derivative evaluation),
  - Surrogate capacity/training (data coverage, architecture mismatch).
- There are **no explicit procedures for non statistical false alarm control**
  - No hierarchical gating where property tests are run only after domain typing & numerical resolution checks.
  - No formal decision rules combining **tolerance bands for numerical artefacts** plus **robustness of violation to mesh/refinement changes**.

**Bottom line:** the field **does not yet interpret failed checks as diagnostic evidence**; failure is almost always “certificate not satisfied,” not “this is a surrogate fault vs regime misspecification vs evaluation artefact.”

## Safety, Reliability, and Credibility Framing

### Numerical analysis “trustworthy AI in numerics”

Several works adopt “trustworthy AI” or “certified ML” language but **stay at the solver level**:

•

Haugen et al. [2] explicitly frame their verification algorithms as **trustworthy AI for PDE solvers** and highlight undecidability of training, arguing for post hoc verification.

- Hillebrecht & Unger [9,22] use “certified machine learning” and “prediction error quantification,” but the safety narrative remains implicit (numerical trust, not system safety).

These works are **natural technical anchors** for an assurance case but do not themselves build it.

### Formal methods style correctness for neural solvers

- BEACONS [15] is conceptually important for safety framing: it integrates **automated theorem proving and formal verification** to **synthesize neural PDE solvers with machine checkable certificates** of convergence, stability, and conservation.
- In spirit, this aligns with **formal methods in safety critical systems** and the properties (stability, conservation, thermodynamic consistency, hyperbolicity) are the same ones that would appear in physical assurance arguments.
- However, BEACONS addresses **design time constructive correctness** not **post hoc V&V of arbitrary surrogates** nor **diagnostic interpretation of failures**.

### Surrogate credibility frameworks in aerodynamics

The only clearly system level, domain specific assurance literature in this set is:

- Kirsch et al. sequence [20,19,18] on **DNN surrogates for aerodynamic coefficients (NACA 0012)**:
  - They use **best practice VVUQ and computational physics model credibility** for the CFD baseline, and extend this logic to the surrogate predictions [20,19].
  - The latest work [18] applies a **SciML adapted Predictive Capability Maturity Model (PCMM)** with elements such as:
    - **Data representation**,
    - **Domain awareness**,
    - **Explainability (SHAP)**,
    - **Model validation**,
    - **Uncertainty quantification**, giving each a **credibility level** and identifying gaps.
- Validation is carried out primarily via:
  - Comparison against **RANS CFD simulations and experimental data** (i.e., trusted oracles),
  - Standard **ML accuracy/UQ metrics**,
  - Interpretability metrics.

Relative to your wish list:

- These papers **do not** use *oracle free* admissibility relations as a primary validation tool; they rely on simulations and experiments as oracles.
- There is **no explicit catalog of conservation/invariance/bounds tests** used as separate evidence.
- They **do** provide an explicit **assurance framing**, vocabulary, and **credibility assessment structure** that could host property based V&V artifacts in future work.

### Connection to PHM/SHM, nuclear, and aerospace safety

Within the provided set:

- There is **no deep treatment** of nuclear I&C or PHM/SHM where neural surrogates are analyzed via property based V&V.
- Ellinas et al. [14] mention **power system dynamics (synchronous generator ODE)** as an application; BEACONS [15] is clearly motivated by **computational physics / plasma / hyperbolic systems**. But safety arguments at the **system level (hazard analysis, safety cases, regulatory considerations)** are only tangentially referenced, if at all.

So, **anchoring to system safety and reliability engineering remains thin** in this corpus and largely decoupled from the rich error certification machinery.

## Implications, Opportunities, and “Where to Build”

### What you can take as available technical infrastructure

For designing your own **pre deployment, oracle free, property based V&V framework** for neural PDE/CFD/operator surrogates, the literature gives you:

- **Mature methods for bounding PDE residuals and solution errors**, with:
  - CROWN/SMT/interval based formal verification [1,2,3,5,14],
  - Variational/functional a posteriori estimators [4,6,7,8,12,16],
  - Semigroup/stability constant based techniques [8,9,21,22],

- Problem specific constructions (e.g., Fokker–Planck [10]).
- **Tools for certified function space norms and local diagnostics** particularly:
  - Certified  $L^p$ ,  $W^{1,p}$ ,  $W^{2,p}$  norms and residual norms [5],
  - Localizable residuals (FoSLS NNs [13], wavelet expansions [6]),
  - Two term error decompositions that isolate numerical vs approximation contributions [12,22].
- **Formal methods integration for correctness properties** from BEACONS [15], showing that:
  - Conservation, stability, hyperbolicity, etc. can be treated as **formal proof obligations** over NN architectures,
  - Automated theorem provers can be interfaced with NN surrogates.

These are **exactly the ingredients** you need to build **reliable evaluators of complex properties** (e.g., conservation residuals, symmetry relations) with **quantified numerical uncertainty**.

## Gaps that are open research directions

The **core gaps** relative to your original goal are:

1. **No explicit, modular catalogues of admissibility properties as offline tests.**
  - You would need to define properties  $P_i(u; \cdot)$  (mass conservation, invariance, bounds, scaling, etc.) and systematically express them as evaluable functionals, possibly with certified numerical error bounds (using tools like [5,4,12]).
1. **No typed admissibility gating or contract system.**
  - None of the works link properties to **regime types** (Re, Ma, geometry, operational mode).
  - You will need to introduce a **type system for inputs and property preconditions**, and then connect property tests to the surrogate's declared domain of validity.
1. **No diagnostic attribution framework for failed property checks.**
  - Existing decompositions (e.g., [22,12]) are promising for separating **numerical vs surrogate contributions**, but no one maps outcomes to categories like:
    - Surrogate defect vs out of domain input vs invalid property assumption vs numerical artefact vs inconclusive.
  - Designing such a **multi axis classification** is yet to be done, but could rest directly on:
    - Certified bounds on numerical evaluation error (via [5,12,22]),
    - Residual to property sensitivity analysis leveraging PDE structure.
1. **Lack of non statistical false alarm control methods.**
  - There is no notion of **false alarm** for property tests beyond trivial “certificate too conservative”; there are no hierarchical test plans or **multi level gating** strategies.
  - Here again you can build on:
    - **Resolution assessments and refinement studies** (e.g., [12,5]) to bound numerical noise before interpreting violations,
    - **Robustness of violations to evaluation variations** (mesh, quadrature, stencils), a concept implicit in numerical - analysis practice but not yet formalized for NN surrogates.
1. **Sparse integration with safety cases and reliability arguments.**
  - Kirsch et al. [18,19,20] provide **credible PCMM style structures** into which your property based V&V artefacts could be slotted as **evidence tables** (e.g., under “domain awareness,” “model validation”).
  - However, you will need to explicitly **extend such frameworks** to include:
    - **Contracts at the surrogate interface**, expressed as admissibility properties,
    - **Traceability** from system level hazards to surrogate level property tests.

## Practical next steps suggested by this literature

- **Combine residual based certification with multi property checks.** Use the existing frameworks [1,2,3,4,5,7,8,9,11,12,13,16,21,22] to ensure that **whenever you test a property**, you know:
  - The surrogate is **within a bounded distance of the PDE solution** in relevant norms;
  - The **numerical evaluation** of your property functional has **certified numerical error  $d$**  (via [5,4,12]).
- **Design property contracts as functional constraints with certified bounds.** Express each admissibility property as a **functional of  $u$**  (and perhaps  $\cdot$ ), then use:
  - Interval/quadrature based techniques [5],
  - Variational dual norms [4], or
  - Localizable residual surrogates [6,13] to compute **guaranteed property residuals plus numerical uncertainty bands**.

- **Introduce typed admissibility and failure attribution layers “above” the current literature.**Starting from the diagnostic decomposition ideas already inherent in [22,12,13], you can:
  - Attach **regime labels and property preconditions** to each test,
  - Develop **hierarchical decision rules** (type check ' numerical adequacy ' property evaluation ' classification),
  - Treat error/residual certifications as **evidence about the test machinery itself**, not just the surrogate.
- **Map certification artefacts into PCMM style credibility structures**For aerospace/nuclear/PHM applications, you can frame:
  - Residual/error certificates [1,2,3,4,6,7,8,9,10,11,12,13,16,21,22] as **verification artefacts** (method correctness),
  - Admissibility tests as **validation artefacts** (does surrogate preserve invariants/system contracts),within a SciML adapted PCMM or safety case similar to [18,19,20].

In summary, the literature you have covers **rigorous, oracle free error certification for neural PDE/ODE surrogates and early credibility assessment for ML surrogates in CFD**, but **stops just short** of the property based, typed, diagnostically rich V&V methodology you are aiming to develop. The existing works supply much of the **numerical and formal backbone** needed to reliably evaluate complex admissibility properties; the **conceptual and organizational layers** (property libraries, typed contracts, diagnostic attribution, false alarm control, and integration into safety cases) remain open territory.

## Categories

### Comparative Overview of Oracle Free Certification and Credibility Frameworks

#### High Level Positioning of the Papers

The references naturally fall into **two main clusters** relative to your topic:

1. **Oracle free error/certificate frameworks for neural PDE/ODE surrogates**Focus: rigorous a posteriori bounds on PDE/ODE residuals or solution errors, using PDE structure but little direct safety/assurance framing.Key refs: [1,2,3,4,5,7,8,9,10,11,12,13,16,21,22,10,11,12,14,15,17].
1. **Credibility / assurance frameworks for ML surrogates in engineering applications**Focus: system level credibility/PCM-M style assessment, largely accuracy and data centric, with limited or no property based admissibility testingKey refs: [18,19,20].

The tables below compare these along dimensions experts typically care about for this topic:**type of surrogate/PDE, kind of oracle free guarantee, role of physics/admissibility properties, treatment of test failure interpretation, and integration into safety/credibility arguments.**

### Error / Residual Certification Frameworks for Neural PDE Solvers

#### Scope, PDE Class, and Surrogate Type

| Ref | Surrogate / Solver Type                                | Governing Problems                                   | Dimension / PDE Class                                 | Parametric / Operator Setting                                 |
|-----|--------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------|
| [1] | Fully connected PINNs                                  | Burgers, Schrödinger, Allen–Cahn, diffusion–sorption | Time dependent PDEs (nonlinear)                       | Single problem PDEs; no explicit operator learning            |
| [2] | Generic NN PDE solvers (incl. PINNs, neural operators) | Abstract linear/nonlinear PDEs                       | General $L^p$ framework; PDE specific residual bounds | Emphasis on general verification, not specific param families |
| [3] | Neural PDE solvers (PINNs, ELMs)                       | Elliptic, parabolic, hyperbolic model classes        | PDE class specific stability assumptions              | Parametric PDEs via operator stability constants              |
| [4] | NN approximations of PDE weak solutions                | Linear elliptic / parabolic (incl. param dependent)  | Space–time Bochner formulations                       | Explicit param dependent diffusion, boundary variation        |
| [5] | Arbitrary NNs, including PINNs                         | General functions; PINN residuals as functionals     | $L^p$ and Sobolev $W^{1,p}$ , $W^{2,p}$               | General; not tied to specific PDE                             |
| [6] | PINNs                                                  | Parameterized PDEs (elliptic)                        | Variational and ultra weak formulations               | Parameterized elliptic PPDEs                                  |
| [7] | PINNs, neural operators, standard regressors           | Elliptic PDEs                                        | Several elliptic prototypical problems                | Explicit parametric PDEs                                      |
| [8] | PINNs                                                  |                                                      |                                                       | Specific flow configuration                                   |

|      |                                            |                                                                                                          |                                              |                                                     |
|------|--------------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------|
|      |                                            | Stokes flow (flow around a cylinder)                                                                     | Stokes equations; semi-group setting         |                                                     |
| [9]  | PINNs                                      | Transport, heat, Navier–Stokes, Klein–Gordon                                                             | Linear PDEs (stated scope)                   | Non parametric examples                             |
| [10] | PINNs for error functions                  | Fokker–Planck PDEs (and related SDEs)                                                                    | Linear PDE class with FP structure           | High dimensional, possibly chaotic SDE induced PDEs |
| [11] | PINNs / NN DE solvers                      | Most linear ODEs, some nonlinear ODEs, first order linear PDEs (2D)                                      | Low dim ODEs + certain 2D PDEs               | Non parametric                                      |
| [12] | NN variational approximations              | Second order elliptic PDEs                                                                               | Variational form; multiple BC treatments     | General elliptic, non parametric                    |
| [13] | FoSLS type NN PDE solvers                  | Linear elliptic, parabolic, hyperbolic                                                                   | First order system least-squares             | General linear PDEs                                 |
| [14] | NN solvers for ODEs/PDEs                   | Burgers, Schrödinger, synchronous generator ODE                                                          | ODE and selected PDE examples                | Non parametric                                      |
| [15] | Formally verified NN PDE solvers (BEACONS) | General PDEs via characteristics decomposition                                                           | Emphasis on hyperbolic / transport type PDEs | Composition of shallow NN solvers as operators      |
| [16] | PINNs                                      | Linear elliptic (primal, mixed, elasticity), parabolic, hyperbolic, Stokes, PDE constrained optimization | General linear PDE classes                   | Problems framed in abstract bilinear form           |
| [21] | PINNs                                      | ODEs (linear and certain nonlinear systems)                                                              | ODEs with monotonic/Lipschitz structure      | Time varying and LTI systems                        |
| [22] | PINNs                                      | ODEs (scalar examples, inverted pendulum)                                                                | ODEs only                                    | Non parametric but out of training inputs allowed   |

**Observation:** almost all PDE oriented works focus on **PINNs for linear PDEs/ODEs** (with a few nonlinear cases) and target **field solutions** rather than operator learning surrogates like FNOs or MeshGraphNets; [2,5,7,13,15,16] are closest to operator or function space viewpoints.

## Type of Oracle Free Guarantee and Use of Physics

| Ref  | Primary Certified Quantity                                                     | Oracle Free Mechanism                                                         | Physics / Admissibility Role                                                                                         |
|------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| [1]  | Upper bound on PDE residual over domain                                        | CROWN formal NN verification to bound nonlinear terms and residual everywhere | Physics enters via <b>PDE residual</b> ; conservation etc. encoded implicitly by PDE, not as separate tests          |
| [2]  | Guaranteed accuracy in $L^p$ norms vs true PDE solution                        | Computable $L^p$ norms of NN and derivatives + PDE specific residual bounds   | Exploits PDE stability theory; <b>no direct conservation/symmetry tests</b> beyond what PDE implies                  |
| [3]  | Deterministic/probabilistic solution error bounds from residuals               | Formal verification (e.g., dReal) of residual bounds + stability constants    | Uses PDE stability (e.g., Lipschitz operator bounds); again, properties are encoded at PDE level                     |
| [4]  | Two sided bounds in problem induced norms (Hilbert)                            | Dual norm of NN residual via Riesz rep. on simpler domains                    | Variational formulation ensures consistency; <b>no explicit property tests</b> like conservation laws                |
| [5]  | Certified upper/lower bounds on $L^p$ and Sobolev norms of NNs (and residuals) | Interval arithmetic + adaptive quadrature over domain                         | Framework is <b>property agnostic</b> ; can certify residual norms but doesn't define physical properties explicitly |
| [6]  | A posteriori upper bound on PINN approximation error                           | Wavelet expansion of residual; scaled coefficient sum e dual norm             | PDE residual is both <b>loss and estimator</b> ; physics fully encoded via PDE formulation                           |
| [7]  | Functional a posteriori error majorants                                        | Auxiliary fields predicted by NN; classical Repin style functionals           | Uses variational functional error estimates; again at <b>PDE level</b> rather than separate admissibility properties |
| [8]  | Rigorous error bounds for PINNs (Stokes)                                       | Semigroup / stability constant based certificates                             | Physics via semigroup theory and Trotter–Kato; no separate property checks                                           |
| [9]  | A posteriori prediction error upper bound                                      | PDE defined PINNs; estimator uses only system characteristics                 | Exploits ODE/PDE system characteristics; <b>no separate conservation/symmetry tests</b>                              |
| [10] | Localized worst case error bounds for Fokker–Planck solutions                  | Recursive error PINNs approximating error functions                           | Uses FP structure; no additional physics properties beyond PDE                                                       |
| [11] |                                                                                |                                                                               | Physics via DE operator only                                                                                         |

|      | Bounds on %u v% via residual based B(x,t) or scalar B                             | Residual $Du = f + r$ ; algorithms from PDE structure                                       |                                                                                                                                                                        |
|------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [12] | Two component a posteriori estimator ( $\cdot, \mu, \bar{A}$ )                    | Variational residual projection + FEM style estimator                                       | Residual and variational structure encode physics; <b>no separate property suite</b>                                                                                   |
| [13] | Least squares residual as norm equivalent error                                   | FoSLS residual equivalent to solution error                                                 | Residual acts as physically grounded indicator; <b>no explicit property catalog</b>                                                                                    |
| [14] | Worst case residual / solution error bounds                                       | Interval / bound propagation (CROWN like) + FD derivatives                                  | PDE/ODE residual is central; correctness phrased in error bounds                                                                                                       |
| [15] | Formal guarantees of convergence, stability, conservation, L extrapolation bounds | Automated theorem proving + method of characteristics; algebraic composition of shallow NNs | <b>Explicitly focuses on conservation/stability</b> but these are enforced as architectural/derivation properties with machine checkable proofs, not as post hoc tests |
| [16] | A priori and a posteriori energy norm error bounds                                | Abstract bilinear form framework; coercivity/continuity                                     | Physics encoded into bilinear forms; not framed as property test library                                                                                               |
| [21] | Two sided error certificates for ODE PINNs                                        | Local strong monotonicity / one sided Lipschitz bounds + residual                           | Contracts are monotonicity/Lipschitz; still solution error centric                                                                                                     |
| [22] | Upper bound on prediction error for ODE defined PINNs                             | ODE structure + residual + numerical integration error decomposition                        | Physics via ODE structure; decomposes PINN vs numerical contributions                                                                                                  |

**Key contrast for your goal:** These works are **strong on oracle free error certification** typically parameterizing guarantees by **PDE residual norms** and stability constants. However:

- **Admissibility properties (conservation, symmetry, monotonicity, bounds)** are almost always treated **implicitly via the PDE itself**, *not* as **separate V&V tests** evaluated post training.
- None of [1,2,3,4,6,7,8,9,10,11,12,13,16,21,22,10,11,12,14] systematically define **property suites** (e.g., mass conservation tests, scaling checks) distinct from the PDE residual, nor do they build **diagnostic attribution frameworks** for violation interpretation in the sense you outlined.
- [15] **BEACONS** does emphasize conservation and stability, but as **hard correctness targets enforced and proved by construction** (architecture + theorem prover), not as independent offline test contracts.

## Interpretation of Failed Checks and Diagnostic/Assurance Orientation

| Ref | Interpretation of Failure                                                             | Distinction Between Model vs Numerical vs Assumption Errors?                                                     | Safety / Assurance Framing                                                              |
|-----|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| [1] | If CROWN cannot certify residual tolerance, NN is uncertified                         | Focus on formal verification feasibility; no explicit decomposition of causes                                    | Framed as "efficient error certification" and trustworthy NN; no system level assurance |
| [2] | NN either passes accuracy verification or is rejected                                 | Emphasizes undecidability of a priori training guarantees; failure means no certificate                          | Discusses "trustworthy AI in numerics"; still at solver level, not system safety        |
| [3] | Large residual bounds 'looser' solution error guarantees                              | Points out role of stability constants; no fine grained fault attribution between surrogate vs PDE assumptions   | Focus on theoretical guarantees; no safety case articulation                            |
| [4] | Wide lower/upper bounds or high residual norms indicate poor approximations           | Discusses computational effectiveness; attributes gap mostly to residual approx. and domain embeddings           | Numerical analysis context; no safety/credibility language                              |
| [5] | Loose function norm bounds indicate insufficient refinement or conservative enclosure | Explicitly attributes conservatism to interval arithmetic and quadrature choices                                 | Positioned as V&V tool; no safety-critical system case                                  |
| [6] | Large wavelet residual based error bound 'insufficient training/approximation         | Focuses on estimator effectiveness; little discussion of test assumption validity                                | Aimed at certified PINNs; no diagnostic taxonomy of failure causes                      |
| [7] | High error majorant signals poor approximation or inadequate auxiliary fields         | Some discussion of training stability; no systematic discrimination of numerical vs modeling vs property failure | Motivates "safer" solvers but not framed as assurance artifact                          |
| [8] | If semigroup based bounds large, certificate is weak/uninformative                    | Mentions practical computability of stability constants; not a diagnostic classification framework               | Numerical analysis; no explicit reliability critical framing                            |
| [9] |                                                                                       | Decomposes error terms (PINN residual, integration error) [22];                                                  | Early steps toward <b>numerical vs model error decomposition</b> still                  |

|      |                                                                                            |                                                                                                         |                                                                                                                            |
|------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
|      | Error bound magnitude used to "certify" or not                                             | distinguishes numerical integration contribution                                                        | not oriented around contract/property admissibility                                                                        |
| [10] | Loose local bounds indicate insufficient error PINN training                               | Focus remains on tightness vs cost; no diagnostics beyond that                                          | Presented as accuracy/bounding tool                                                                                        |
| [11] | Different algorithms offer tighter or faster bounds                                        | Some complexity vs tightness analysis; no classification of violation causes                            | Error bound centric                                                                                                        |
| [12] | Large $\sigma$ / $\mu$ / $\hat{\sigma}$ highlights regions or aspects needing refinement   | Discussion on quadrature effects and localized phenomena; acknowledges quadrature errors                | Offers a clear <b>numerical vs approximation perspective</b> (e.g., quadrature adequacy) but not full admissibility gating |
| [13] | Large FoSLS residual indicates local "out of equilibrium" zones                            | Designed for adaptivity; explicitly interprets residual as localization indicator                       | Still targeted at training/solver adaptivity, not safety certification                                                     |
| [14] | Inability to bound residual/solutions within tolerance ' NN fails correctness verification | Introduces parallel branching and FGSM style attacks; failure attributed to network/verification limits | Motivated partly by power system safety, but interpretation remains "pass/fail" not typed contracts                        |
| [15] | If formal proof fails, NN solver not accepted; violations correspond to proof obligations  | Interpretation dominated by theorem prover failures, not by run time property tests                     | Strong notion of <b>formal assurance</b> , but again by construction rather than offline property tests                    |
| [16] | Error estimator behavior informs training choices and problem formulations                 | Discusses quadrature and sampling errors; not structured as contract failure attribution                | Theoretical error analysis emphasis                                                                                        |
| [21] | Upper/lower certificates classify whether PINN satisfies error contracts                   | Discusses limitations (e.g., soft vs hard IC enforcement, monotone vs conservative systems) explicitly  | Comes closest in ODE context to <b>component level contract reasoning</b> , though still error centric                     |
| [22] | Overestimated bounds or trivial lower bounds limit certificate usefulness                  | Clearly separates PINN residual and numerical integration contributions; suggests mitigation strategies | Explicit mention of limitations; but no higher level assurance framework                                                   |

Across these works, **failure is almost always interpreted as "insufficient accuracy / unverifiable error bound"**, rather than as a structured diagnostic:

- There is **little notion of typed admissibility contracts**: properties are not partitioned by regime, geometry, or operational mode.
- Tests rarely distinguish **invalid property preconditions** (e.g., conservation law inapplicable) from **surrogate faults**.
- Only a few works, notably [12] and [22], **explicitly decompose numerical vs approximation contributions** to the certified quantity, which is a building block for your envisioned diagnostic attribution (but still not fully developed).

## Role of Admissibility / Physics Properties vs Pure Error Bounds

### Property Orientation vs Residual/Norm Orientation

To your specific interest in **oracle free, property based validation (conservation, invariance, monotonicity, bounds, etc.) used as tests rather than architectural constraints**, the comparison is as follows:

- **Predominant pattern**: almost all certification works treat **the PDE residual and its norms** as the **sole primary property**:
  - Residual H "physics consistency" because the PDE encodes conservation/invariance implicitly.
  - No explicit **catalogues of secondary properties** such as:
    - Global mass/energy balance independent of PDE residual norm;
    - Symmetry or invariance checks under transformations;
    - Monotonicity or bound constraints formulated as independent contracts.
- **[15] BEACONS** is unique in explicitly emphasizing properties like **conservation, stability, convergence** and aiming for **machine checkable certificates** of these. However:
  - These properties are **proven as part of the architecture design and synthesis**, not used as independent offline V&V tests applied to a generic surrogate.
  - It therefore aligns more with **"by construction" correctness** than with **post hoc property testing**

In summary, **none of the technical certification papers in [1,2,3,4,6,7,8,9,10,11,12,13,14,15,16,21,22] develop the kind of multi property admissibility test suite you described** nor do they emphasize **oracle free checks of conservation/symmetry/monotonicity/etc. as standalone V&V artifacts**. Instead, they converge on **residual based error certification** as the main mechanism.

## Credibility and Assurance Frameworks for Engineering Surrogates

### SciML / CFD Surrogate Credibility in Aerodynamics

| Ref  | Application                                         | Surrogate Type                    | Credibility / Assurance Framework                                                                                                                | Use of Physics Based Properties in V&V                                                                                                                                                                                                | Treatment of Failure / Evidence                                                                                                                         |
|------|-----------------------------------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| [18] | NACA 0012 airfoil aerodynamic coefficients          | DNN surrogate trained on RANS CFD | SciML adapted Predictive Capability Maturity Model (PCMM): evaluates data representation, domain awareness, explainability, model validation, UQ | Uses SHAP interpretability, comparisons to experimental and simulation data, and numerical/surrogate run to run uncertainty; no mention (in excerpt) of explicit conservation/invariance/bound checks as <b>oracle free contracts</b> | Assigns credibility levels per PCMM element; identifies gaps; focuses on <b>evidence structure and maturity</b> , not on detailed property based gating |
| [19] | Surrogate for subsonic turbulent airfoil flow       | ANN surrogate                     | "Wholistic" credibility assessment combining computational model V&V elements with ML accuracy/UQ                                                | Emphasis on best practices for validation (likely via comparison to CFD/experiments), accuracy metrics, UQ; no explicit oracle free property test suite                                                                               | Intended to guide future surrogate use; positions credibility assessment as combination of classic V&V and ML evaluation                                |
| [20] | DNN surrogate for lift/drag prediction on NACA 0012 | DNN                               | Best practice credibility analysis for CFD and surrogate predictions                                                                             | Validates against experimental and CFD baselines; discusses credibility of both solver and surrogate; again, <b>comparison with trusted oracles</b> appears central                                                                   | Draws conclusions for future surrogate use; underlines need for standard surrogate - credibility practice                                               |

Relative to your desiderata:

- These works **explicitly address credibility, trust, and evidence structure** for surrogates in engineering contexts (CFD/aerodynamics), which is directly relevant as **framing for assurance**.
- However, their **validation evidence appears dominated by**:
  - **Comparison with RANS CFD and experiments** as oracles;
  - **Standard ML accuracy and UQ metrics**;
  - **High level PCMM style dimension scores**(data, domain awareness, etc.).
- They do **not** (in the provided descriptions) introduce:
  - **Oracle free admissibility tests**(e.g., conservation tests independent of CFD oracle);
  - **Typed admissibility gating** or formal property contracts;
  - **Diagnostic classification of property test failures** (e.g., surrogate vs assumption vs numerical artifact).

### Contrasting Technical Certification and System Level Credibility

| Aspect                    | Technical Certification Papers<br>[1,2,3,4,6,7,8,9,10,11,12,13,14,15,16,21,22,10,11,12,13,14,5,7,8,11,12]              | Credibility Papers [18,19,20]<br>[1,2,3,4,6,7,8,9,10,11,12,13,14,5,7,8,11,12]                                            |
|---------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Primary goal              | Rigorous oracle free <b>error/residual bounds</b> for neural PDE/ODE solvers                                           | Structured <b>credibility assessment</b> of ML surrogates embedded in engineering workflows                              |
| Evidence type             | Mathematical stability, residual bounds, function norm certificates                                                    | PCMM like scores, validation vs CFD/experiments, accuracy/UQ metrics, interpretability (SHAP)                            |
| Physics properties        | Encoded almost exclusively via <b>PDE residuals and variational forms</b> ; conservation/invariance rarely singled out | Physics used mainly through <b>high fidelity CFD and experimental oracles</b> ; not via explicit admissibility relations |
| Runtime vs pre deployment | Entirely <b>offline, pre deployment</b> error/certification procedures                                                 | Also primarily offline assessments, targeted at <b>model deployment decisions</b>                                        |
| Interpretation of failure | Failure = unverifiable/tight error bound; occasionally numerical vs model error separation                             | Failure = inadequate credibility levels or poor accuracy/UQ vs oracles; no property test failure taxonomy                |
| Safety/assurance linkage  |                                                                                                                        | Explicit credibility framing in <b>CFD design and safety related engineering contexts</b> but with                       |

|                                                                           |                                                             |
|---------------------------------------------------------------------------|-------------------------------------------------------------|
| Implicit "trustworthy numerics"; largely absent system level safety cases | limited technical depth on oracle free admissibility checks |
|---------------------------------------------------------------------------|-------------------------------------------------------------|

## Synthesis: What Is Unique and What Is Missing Relative to Your Topic

### Uniquely Relevant Elements Across the Papers

Even though none of the works fully realize the **admissibility test + diagnostic attribution + safety case** workflow you described, several **partial ingredients** emerge:

- **Formal NN verification for PDE residuals:** [1,2,3,5,14,15] use **CROWN style bound propagation, interval arithmetic, or theorem proving** to obtain domain wide guarantees on either residuals or function norms, which is directly relevant to designing **oracle free property checkers** (even if they presently focus only on residuals).
- **Error/residual to solution bounds in PDE settings** [3,4,6,7,8,9,10,11,12,13,16,21,22] provide concrete **residual to solution error mappings** and **a posteriori estimators**, which could underpin **tolerance based admissibility checks** for properties derived from PDE solutions.
- **Explicit numerical vs model error decomposition:** [22] decomposes total error into PINN residual vs numerical integration error; [12] decomposes error into residual projection and complementary estimator; [13] and [5] emphasize quadrature and interval arithmetic contributions. These are directly useful for **distinguishing numerical artifacts from surrogate faults**.
- **Component level contract flavor (in ODEs)** [21] introduces **upper and lower "certificates"** under monotonicity/Lipschitz conditions and explicitly discusses situations where bounds degenerate or lose informativeness, which resembles **contract style reasoning** in a limited ODE context.
- **Formal assurance tooling:** [15] demonstrates that **automated theorem provers and formal methods** can be integrated with neural PDE solvers to generate **machine checkable correctness certificates** something that could be extended in principle to **property contracts** beyond residuals.
- **System level credibility frameworks in CFD** [18,19,20] anchor the discussion in **engineering credibility** (SciML adapted PCMM, holistic surrogate assessment), giving a template for how property based V&V artifacts might be integrated into **broader assurance cases**.

### Gaps Relative to Oracle Free Property Based V&V with Diagnostic Attribution

Across all references:

- **Property suites beyond residuals are largely absent:** explicit tests of **conservation, invariance, scaling, monotonicity, bounds, dimensional consistency** as **post hoc, oracle free checks** are not systematically developed.
- **Typed admissibility / contract systems** are not present:
  - None of the works formalize **input typing based on regime (e.g., Re, Ma, geometry)** and then **guard property applicability** based on those types.
  - Consequently, **property inapplicability vs surrogate failure** is not distinguished.
- **Non statistical false alarm control** (in your sense) is missing:
  - There is almost no discussion of **false alarms arising from misapplied properties or numerical noise**.
  - Decision rules are mostly **scalar thresholds on residual/error bounds** rather than **hierarchical gating** (type ' numerical adequacy ' property test).
- **Diagnostic attribution frameworks are embryonic at best:**
  - Only [22,12], and to some degree [13] explicitly partition error contributions into **approximation vs numerical** terms.
  - No paper develops a **multi axis classification** (domain of validity vs physics assumption vs numerical evaluation vs surrogate model) as you outlined.
- **Safety critical integration is high level**
  - [18,19,20] provide **credibility framing** but not **technical property based V&V methods** beyond comparisons to oracles.
  - The certification papers rarely, if ever, connect their certificates to **formal safety cases, hazard analysis, or regulatory acceptance**.

For an expert researcher, this comparative picture means:

- The field has **matured substantially on oracle free error/residual certification** for neural PDE/ODE solvers, especially in **linear (and some nonlinear) PINN based settings**
- However, **explicit property based offline V&V** in the sense of **independent admissibility tests and diagnostic attribution** is largely **unexplored** in the current literature, representing a clear research opportunity at the intersection of **numerical analysis, formal methods, and system safety**.

## Timeline

## Early certified error bounds for PINNs (2022–early 2023)

### Foundational ODE-focused certification

- **2022 – Hillebrecht & Unger’s ODE PINN certificates** The first clearly articulated “certified ML” line in this corpus is the ODE-focused a posteriori error bound for PINNs by **Hillebrecht & Unger** [22].
  - They derive a **rigorous oracle free upper bound** on the prediction error of ODE-defined PINNs, based solely on the governing equations and the network’s residual, with no access to the true solution.
  - The paper explicitly decomposes error contributions (PINN residual, numerical integration error, etc.) and gives **explicit bounds on numerical-integration error**, including subinterval requirements for trapezoidal rule [22].
  - A secondary “Error NN” is introduced to approximate the error estimator efficiently, foreshadowing later “dual” or auxiliary networks learning error indicators. This work effectively seeds the notion of **certified PINNs** and is explicitly framed as “certified machine learning.”
- **2022 – PDE focused extension of Hillebrecht & Unger** Their 2022 IEEE TNNLS paper [9] extends the certification philosophy to **PDE-defined PINNs** (linear PDEs), deriving rigorous a posteriori upper bounds for PDE problems (transport, heat, Navier–Stokes, Klein–Gordon).
  - Unlike [22], which is ODE-only, [9] provides a first PDE-level instantiation of **oracle free prediction-error certificates** for PINNs, still centered on residual-based analysis and dynamical-system properties.
  - This moves the field from toy ODE examples toward **fluid and wave PDEs**, with explicit emphasis on “certified machine learning” and “prediction error quantification.”

These two works [22,9] mark a first milestone: they **reframe PINN accuracy as a certified quantity** with a posteriori bounds, rather than just empirical validation, and they explicitly use **error certification language** that later papers adopt and generalize.

### Early wavelet-based and variational approaches

- **2022 – Wavelet-certified PINNs** In parallel, the wavelet-based certified PINN [6] develops an adaptive-wavelet representation of the PDE residual to approximate the dual norm of the residual and derive upper bounds.
  - The wavelet expansion coefficients are used as both **training loss and explicit error estimator**, connecting functional-analytic residual norms with PINN implementations.
  - This anticipates later work on dual-norm and functional a posteriori estimators, and it contributes a different numerical technology (wavelets vs. quadrature / CROWN / interval arithmetic).

Taken together, by ~2022 the field has:

- ODE and linear-PDE **a posteriori error certificates** for PINNs [22,9].
- Early **functional residual** tools via wavelets [6].
- The central idea: **certification via residual-based bounds** without oracle access is now clearly established.

## Expansion to PDEs and residual-based error bounding (2023–2024)

### Residual-to-error frameworks and generalization bounds

- **2023 – Residual-based error bounds (Liu et al.)** Liu et al. [11] provide residual-based, oracle-free error bounds for neural-network solutions of ODEs and certain PDEs, focusing on deriving explicit **spatio-temporal bounds**  $B(x, t)$  or global bounds  $B$  such that  $\|u - u_h\| \leq B$ .
  - They explore multiple algorithms with different tightness vs runtime trade-offs, and rely entirely on **equation structure plus residuals**, not on training assumptions [11].
  - Method-of-manufactured-solutions examples are used to verify tightness, tying the emerging certification line back into classical verification practice.
- **2024–2026 – “Residual-to-error” certification** Mukherjee et al. [3] unify the residual-to-error philosophy, proving deterministic and probabilistic **generalization/convergence results**: vanishing residuals (plus boundary/initial errors) imply convergence in solution space under stability assumptions.
  - They explicitly propose a **four-step certification pipeline**: training; formal residual verification (e.g., via dReal); stability constant computation; and translation into solution-error guarantees [3].
  - This framework connects **formal methods** (SMT-based residual verification) with classical PDE stability theory, representing a conceptual milestone: residual bounds become the primary “property” whose correctness is formally certified and then translated into error bounds.

These works consolidate the idea that **rigorous residual control** is sufficient to certify surrogate accuracy, bridging residual-based certification with **PDE stability and generalization theory**.

### A posteriori PINN error analysis with semigroup methods

- **2022–2025 – Semigroup and stability-based certificates** The same Hillebrecht & Unger line evolves with a rigorous PINN a posteriori PDE estimator [9] and a later semigroup-based, practically computable framework [8] for Stokes flow.
  - [8] modifies earlier semigroup input-to-state-stability bounds to make them numerically tractable, including strategies to approximate the relevant stability constants.
  - This marks a shift from purely theoretical error bounds toward **algorithms that can be used in realistic flow problems**, albeit still in a numerical-analysis context.

### Error estimation as part of a unified PINN error framework

- **2024 – Unified error analysis (Zeinhofer et al.)** Zeinhofer et al. [16] give an abstract bilinear-form framework for PINNs for a wide class of linear PDEs, combining **a priori and a posteriori** error analysis.
  - They adapt Cea-type arguments to NN classes and derive coercivity/continuity conditions yielding sharp error bounds, with constants independent of network parameterization [16].
  - While [16] is more theory-focused and less explicitly “certification language,” it shows a growing **confluence between numerical analysis and neural-PDE surrogates**, embedding PINNs firmly within classical PDE error-analysis paradigms.

## Emergence of certified function-space norms and formal NN-PDE verification (2023–2026)

### Function-space norm certification via interval and quadrature methods

- **2023–2026 – Function-space norm certification (Gründler et al.)** Gründler et al. [5] focus explicitly on **certified computation of  $L^p$  and Sobolev norms** of neural networks, including derivatives.
  - Using interval arithmetic over box partitions, adaptive refinement (Dörfler marking), and quadrature, they obtain **certified bounds on integral functionals** and show how this yields certified bounds on **PINN residual norms** [5].
  - This work is a key milestone in shifting from pointwise bounds to **global function-space certificates**, bridging NN verification with PDE norms.
- **2023 – CROWN extension and residual certification (Eiras et al.)** Eiras et al. [1] tailor the CROWN verification framework to PINNs, introducing **-CROWN** to compute rigorous bounds on nonlinear PDE residuals over entire spatio-temporal domains.
  - They derive **global correctness conditions** based on worst-case residual norms and adapt NN verification techniques to PDE residuals.
  - Compared to [5], which builds new interval/quadrature machinery, [1] leverages the CROWN lineage to bring **formal NN-verification techniques firmly into PINN/PDE residual certification**.

Together, [1,5] signal a trend: **general-purpose NN verification and interval methods are being repurposed for PDE residual certification**, not just for classification or control networks.

### General verification algorithms for neural PDE solvers

- **2025 – Verification algorithms for neural PDE solvers (Haugen et al.)** Haugen et al. [2] propose a posteriori verification algorithms that compute **rigorous estimates of  $L^p$  norms of networks and their derivatives**, then combine them with PDE-specific residual bounds to certify accuracy.
  - This explicitly targets **trustworthy AI in numerics**, acknowledging the undecidability of training and focusing on **computable post hoc verification** independent of training [2].
  - The emphasis is on **computable, problem-independent verification procedures** for PINNs and neural operators.
- **2024 – Correctness verification via CROWN + adversarial search (Ellinas et al.)** Ellinas et al. [14] propose a correctness-verification framework using finite-difference derivatives, CROWN-style bound propagation, and gradient-based attacks to bound NN outputs and residuals for differential equations.
  - Their branching algorithm delivers **worst-case error guarantees** and explores **adversarial-style searches** over PDE solution domains [14].
  - This marks a convergence of **adversarial robustness techniques** and **PDE residual verification**.

These works collectively show a transition from **PINN-specific residual analysis** to **general NN-PDE verification toolchains** that borrow heavily from NN verification and adversarial robustness communities.

## Integration with variational and functional a posteriori frameworks (2023–2025)

### Variational a posteriori estimators and dual-decomposition

- **2025 – A posteriori analysis of NN PDE approximations (Führer & Rojas)** Führer & Rojas [12] formulate error estimation in variational form, decomposing total error into a **MINRES projection term**  $\mu$  (lower bound) and a FEM-style estimator  $\hat{\mu}$ , with a computable upper bound  $\hat{A}$  [12].

- They emphasize quadrature accuracy and propose adaptive quadrature for robustness near localized features, deepening the **numerical-analysis rigor** of NN error estimation.
- This reinforces the view of NN surrogates as **approximate finite-element-like objects**, for which standard a posteriori techniques can be adapted.
- **2023–2024 – Functional a posteriori error estimates as training losses (Fanaskov et al.)** Fanaskov et al. [7] build NN architectures that **predict auxiliary fields** enabling classical functional a posteriori majorants; the majorant is computable as a cheap postprocessing step.
  - Their key methodological twist is to embed these error estimates **directly into the loss function**, linking training to later certification while still providing post-hoc upper bounds [7].
  - They show this works across multiple architectures (PINNs, neural operators, etc.), pushing the certification philosophy beyond PINNs.

## FoSLS-based residuals and localized diagnostics

- **2024 – First-Order System Least Squares Neural Networks (FoSLS NNs)** Opschoor et al. [13] develop FoSLS-based NN PDE solvers where the **L<sup>2</sup> least-squares residual** functions as both training loss and an a posteriori indicator.
  - They leverage de Rham-compatible NN parameterizations and derive equivalence between LSQ residuals and error norms [13].
  - Crucially, the residual is **additive and localizable**, creating **local “out-of-equilibrium” indicators** useful for offline diagnostics and adaptive refinement.

Overall, these works embed neural PDE surrogates into the **standard a posteriori toolbox of variational and functional error estimation**, creating a bridge from pure residual bounds to **structured error decompositions and localization**.

## Two-sided error certificates and lower bounds (2022–2026)

### Two-sided and lower-bound certificates

- Hillebrecht & Unger's ODE and PDE works [22,9] provide **upper bounds**; later work seeks **two-sided bounds**.
- **2026 – Two-sided PINN error bounds (Huseynov et al.)** Huseynov et al. [21] derive **upper and lower a posteriori error bounds** for ODE PINNs under monotonicity and one-sided Lipschitz assumptions.
  - They analyze the effect of soft vs hard enforcement of initial conditions and introduce **signed-residual finite-probe certificates** to recover informative lower bounds [21].
  - They also suggest **certificate-informed training**, where the upper certificate regularizes training while lower certificates serve as post-training diagnostics.

This evolution from purely upper-bound certificates [22,9,11] to **two-sided certificates** [21,12] marks a refinement phase focused on **tightness and informativeness** of error bounds and highlights limitations in highly conservative, one-sided bounds.

## Formal verification and algebraic composition of neural PDE solvers (2026)

### BEACONS: formally verified composable neural solvers

- **2026 – BEACONS (Gorard et al.)** BEACONS [15] is a conceptual milestone: it treats neural networks as **general numerical methods** and uses an automated theorem prover plus code generator to synthesize **formally verified neural PDE solvers**.
  - They claim provable convergence, stability, and conservation, along with **machine-checkable certificates**, e.g., for thermodynamic consistency and hyperbolicity preservation [15].
  - The approach exploits analytical properties (e.g., method-of-characteristics insights) and decomposes PDE solutions into **algebraic compositions of shallow neural approximators**, managing L extrapolation errors through algebraic composition.
  - This sits at the intersection of **formal methods** and numerical PDEs, going beyond residual bounding to **construct neural architectures with formally proved properties**.

BEACONS thus extends the field beyond post-hoc verification to **design-time synthesis of provably correct neural PDE solvers**, echoing earlier formally verified classical solvers but now in a neural context.

## Error-certification for specific PDE classes (e.g., Fokker–Planck) (2024–2025)

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**2024 – Fokker–Planck error bounds (Kong et al.)** Kong et al. [10] tailor error-certification methodology to Fokker–Planck PDEs, learning a recursive series of error functions (error-PINNs) and proving convergence, leading to **localized error bounds** in time and space.

- They emphasize **worst-case localized error** rather than global norms and demonstrate the approach in high-dimensional, chaotic settings [10].
- This is emblematic of a trend toward **problem-specific certification frameworks**, complementing general residual-to-error theories.

## Credibility and assurance frameworks for ML surrogates in CFD (2024–2026)

Although most of the above works live in a **numerical-analysis / verification** paradigm, a separate line focuses on **credibility and assurance in engineering CFD**.

### SciML-adapted PCMM and credibility assessments

- **2024–2026 – Kirsch et al.’s CFD surrogate credibility line** The sequence [20,19,18] by Kirsch, Rider, Fathi, and coauthors develops a **credibility-assessment methodology** for ML-based aerodynamic surrogates (DNNs predicting lift/drag for NACA 0012):
  - [20] applies best practices in **credibility of computational physics models** plus generic ML accuracy and uncertainty evaluation to a DNN surrogate for lift/drag prediction.
  - [19] extends this to a “**wholistic**” **credibility assessment**, proposing a unified approach for surrogate credence in aerodynamic design contexts.
  - [18] explicitly applies a **SciML-adapted Predictive Capability Maturity Model (PCMM)** to a DNN surrogate, assessing **data representation, domain awareness, explainability, validation, and UQ** within a structured credibility framework [18].
  - They use SHAP for interpretability and combine **simulation and experimental data** in validation.

While these papers do not develop the **oracle-free property-based admissibility checks** seen in the PINN-certification literature, they do represent early efforts to **import established V&V and credibility frameworks** into the ML-surrogate context, particularly in aerospace CFD. They provide important **assurance framing and terminology** (e.g., maturity levels, credibility elements) that could be married with property-based V&V in future work.

## Overview of key research clusters and contributors

### Hillebrecht & Unger cluster (certified PINNs via semigroup and residual analysis)

- Core works: [22] (ODE certified PINNs), [9] (PDE-certified PINNs), [8] (semigroup-based practical certificates), and contributions cited in later works [2,21].
- Distinctives:
  - Early and persistent use of “**certification**” language for PINNs.
  - Blend of **semigroup theory, input-to-state stability**, and practical error-bounding algorithms.
- Impact:
  - Their lineage is heavily cited in later error-certification and residual-based works [2,7,8,21], effectively establishing a **canonical reference for a posteriori error bounds** in PINNs.

### Urban / Ernst / Führer line (variational a posteriori NN error analysis)

- Core works: wavelet-certified PINNs [6]; a posteriori certification via variational dual norms [4]; and Führer & Rojas’s a posteriori analysis [12].
- Distinctives:
  - Systematic **variational and Hilbert-space embedding** of NN approximations, leveraging dual norms, Riesz representations, and FEM-style estimators [4,12].
  - Emphasis on **geometrically simpler domains** and efficient dual-norm computation [4].
- Impact:
  - Strengthens the connection between **classical finite-element a posteriori analysis** and NN-based PDE surrogates; provides a toolbox for error decomposition and lower/upper bounds.

### Residual-to-error / function-space verification cluster (Eiras, Gründler, Haugen, Mukherjee, Liu)

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Eiras et al.'s CROWN-based PINN verification [1], Gründler et al.'s function-space norm certification [5], Haugen et al.'s general NN PDE verification [2], Liu et al.'s residual-based bounds [11], and Mukherjee et al.'s residual-to-error framework [3] form a cluster focused on **residual certification and its translation to error guarantees**.

- Distinctives:
  - Heavy use of **formal NN verification tools** (CROWN, SMT/dReal) and **interval / adaptive quadrature** techniques.
  - Emphasis on **rigorous  $L^p$  and Sobolev bounds**, and **architecture-independent verification algorithms** [2,5].
- Impact:
  - This cluster is pivotal for moving the field toward **general-purpose NN-PDE verification toolchains**, decoupled from specific PINN architectures.

### Functional a posteriori / auxiliary-field cluster (Fanaskov & Oseledets)

- Fanaskov et al. [7] integrate functional a posteriori error majorants directly into NN training and postprocessing.
- Distinctives:
  - View error majorants as **both training objectives and certification quantities**, bridging design-time and post hoc V&V.
- Impact:
  - This cluster highlights **auxiliary field prediction** as a mechanism to obtain richer error diagnostics and may foreshadow more elaborate **multi-head networks** that jointly predict solution and error indicators.

### Formal methods and algebraic composition cluster (BEACONS)

- BEACONS [15] is associated with Gorard, Hakim, Juno and earlier work on formally verified classical solvers.
- Distinctives:
  - Use of **automated theorem proving** and **machine-checkable certificates** for neural solvers, focusing on conservation, stability, and hyperbolicity.
  - Structural decomposition of PDEs into **algebraic compositions of proven building blocks**.
- Impact:
  - Introduces a **formal-methods perspective** into neural PDE solvers, suggesting future integration with safety-case and regulatory artifacts.

### Assurance and credibility frameworks cluster (Kirsch, Rider, Fathi)

- The sequence [20,19,18] provides sustained work on **credibility assessment** of ML surrogates in aerodynamics, using PCMM and VVUQ practices.
- Distinctives:
  - Focus on **structured credibility assessment** (data, domain awareness, explainability, validation, UQ) rather than purely numerical error bounds.
- Impact:
  - Offers a **bridge to system safety and regulatory discourse**, a necessary complement to technical certification for eventual deployment in critical systems.

## Cross-cutting trends and shifts in focus

### From ODE toy problems to realistic PDEs and flows

- Early certification efforts [22] began with ODEs and simple PDEs; by ~2024–2026, we see:
  - **Stokes flow and flow around a cylinder** [8].
  - **Fokker–Planck / high-dimensional stochastic dynamics** [10].
  - General PDE solver verification frameworks applicable to **elliptic, parabolic, hyperbolic, and Stokes problems** [2,3,16].

This reflects a steady **expansion in application scope**, moving toward problems closer to real CFD/physics usage.

### From single-sided, highly conservative bounds to two-sided and localized diagnostics

- Early work mostly delivered **global upper bounds** (often conservative). Later work focuses on:
  - **Two-sided bounds** and lower certificates [12,21].
  - **Localized diagnostics**, either via wavelet coefficients [6], FoSLS residual localization [13], or localized error-PINNs [10].

- The significance is a shift from “any bound is good” to “bounds must be **informative, local, and usable** in practice,” especially for model refinement and property diagnostics.

### From PINN-specific error analysis to architecture-agnostic verification

- PINNs are still a central case, but later works emphasize:
  - Applicability to **neural operators and other surrogates** [2,7].
  - Verification algorithms **independent of training procedures and architectures** [2,4,5].
- This trend is crucial for your target domain (MeshGraphNets, FNOs, etc.): the verification machinery is increasingly **operator/surrogate-agnostic**, enabling broader reuse.

### From pure error metrics to formalized verification and conformal UQ

- There is a progression from **error as a scalar metric** to:
  - **Formal verification** of residuals via CROWN/SMT [1,3,14].
  - **Certified integral norms and Sobolev norms** [5].
  - **Conformal prediction with physics residuals as nonconformity scores (PRE)** [17], which introduces a **physics-informed UQ** lens with oracle-free calibration.
- While [17] is more focused on UQ than on property-testing per se, it continues the trend of making **residuals central to trust assessment**, including statistical notions of coverage.

### Lag between error certification and property-based admissibility testing

- Most works in this corpus treat **PDE residuals as the primary property** being certified.
- There is minimal explicit development of:
  - **Independent physical invariants** (e.g., energy, momentum) as test properties not enforced in training.
  - **Typed admissibility gating or contract-based property applicability**.
  - **Diagnostic attribution of failed checks** (surrogate vs assumption vs discretization).
- This gap suggests that the field has **matured around error certification** but has yet to fully develop **property-based V&V and interpretive frameworks** that align with system safety practices.

## Implications and future directions indicated by the timeline

### 1. Consolidation of rigorous, oracle-free error certification

- Between [22,9,6,11,1,5,2,3,12,21], the community now has multiple, theoretically sound, and computable ways to obtain **error guarantees** for neural PDE surrogates, especially PINNs.
- Any future property-based V&V framework can treat these error certificates as baseline **verification artefacts**, ensuring that any property testing is conducted on surrogates with known error envelopes.

### 1. Growing alignment with classical numerical V&V

- Variational frameworks [4,12,16], functional a posteriori theory [7], and semigroup/stability tools [8,9] mean neural surrogates are increasingly treated as **just another numerical method**.
- This alignment is promising for integration with **traditional V&V standards** in nuclear/aerospace domains.

### 1. Early but incomplete integration with system-level assurance

- The Kirsch et al. credibility line [18,19,20] and the formal-methods BEACONS framework [15] show **nascent attempts to connect error certification and credibility to system-level assurance arguments**.
- However, there is still little explicit work on:
  - **Admissibility gating**, typed input regimes, or contract-style property applicability.
  - **Interpretation of failed property checks**, including non-statistical false alarm control and attribution among surrogate, numerical, and assumption errors.

### 1. Opportunity for property-based V&V and diagnostics

- The technical foundations—formal residual certification, function norms, variational decompositions, and localized error diagnostics—are now in place.
- What is largely missing, and thus an opportunity for future research aligned with your goal, is:
  - Systematic **design of test suites of independent physical admissibility properties** (conservation, invariance, monotonicity, bounds) as **offline contracts**.
  - **Typed admissibility and contract frameworks** for which properties are applicable under which regimes.

- **Diagnostic pipelines** that map property-test failures to categories (surrogate defect vs property inapplicability vs numerical artifact), integrating numerical a posteriori diagnostics with physical assumption tracking.

### 1. Emerging cross-disciplinary collaborations

- The clusters identified—numerical analysts (Urban, Urban's students, Oseledets), NN-verification researchers (Eiras, Gründler, Haugen), and system-credibility practitioners (Kirsch, Rider)—have largely worked in parallel.
- Future progress on property-based, offline V&V in safety-critical contexts will likely come from **cross-pollination**:
  - Using **functional/a posteriori tools** to bound numerical artefacts in property evaluation.
  - Feeding **residual-based certificates** into **credibility models like PCMM**.
  - Leveraging **formal methods (BEACONS, CROWN)** to provide machine-checkable property contracts beyond residuals.

Overall, the timeline shows a **rapid maturation (2022–2026)** from isolated PINN error bounds to a rich ecosystem of a posteriori certification, functional norms, formal verification, and early credibility frameworks. The next phase, not yet fully realized in this literature, is to **translate these tools into structured, property-based V&V with diagnostic interpretation**, especially for neural operators and operator-learning surrogates deployed in safety-critical engineering systems.

## Foundational Work

### Which papers form the foundational references on this topic?

The below table shows the resources that are most often cited by the relevant papers on this topic. This is measured by the **reference rate**, which is the fraction of relevant papers that cite a resource. Use this table to determine the most important core papers to be familiar with if you want to deeply understand this topic. Some of these core papers may not be directly relevant to the topic, but provide important context.

| Ref.  | Reference Rate | Title                                                                                                                                                   | Cited By These Relevant Papers |
|-------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| [118] | 0.39           | Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations | [1, 2, 4, 5, 9, 11, 15, 17]    |
| [22]  | 0.25           | Certified machine learning: A posteriori error estimation for physics-informed neural networks                                                          | [1, 4, 5, 9, 21]               |
| [47]  | 0.20           | Estimates on the generalization error of Physics Informed Neural Networks (PINNs) for approximating PDEs                                                | [1, 2, 4]                      |
| [38]  | 0.17           | A Unified Framework for the Error Analysis of Physics-Informed Neural Networks                                                                          | [2, 4]                         |
| [57]  | 0.16           | Generic bounds on the approximation error for physics-informed (and) operator learning                                                                  | [1, 9, 11, 21]                 |
| [39]  | 0.16           | Error analysis for physics-informed neural networks (PINNs) approximating Kolmogorov PDEs                                                               | [2, 4, 11]                     |
| [94]  | 0.16           | Artificial neural networks for solving ordinary and partial differential equations                                                                      | [2, 4, 11]                     |
| [100] | 0.16           | Neural-network methods for boundary value problems with irregular boundaries                                                                            | [2, 4, 11]                     |
| [141] | 0.15           | DGM: A deep learning algorithm for solving partial differential equations                                                                               | [4, 5, 11]                     |
| [124] | 0.13           | Machine learning solutions looking for PDE problems                                                                                                     | [2]                            |
| [1]   | 0.12           | Efficient Error Certification for Physics-Informed Neural Networks                                                                                      | [3, 15, 21]                    |
| [9]   | 0.12           | Rigorous a Posteriori Error Bounds for PDE-Defined PINNs                                                                                                | [2, 8]                         |
| [79]  | 0.12           | Hidden physics models: Machine learning of nonlinear partial differential equations                                                                     | [2, 4]                         |
| [105] | 0.12           | Neural-network-based approximations for solving partial differential equations                                                                          | [2, 4]                         |
| [4]   | 0.12           | A posteriori certification for neural network approximations to PDEs                                                                                    | [12, 15]                       |
| [142] | 0.12           | Quasi-Optimal Least Squares: Inhomogeneous boundary conditions, and application with machine learning                                                   | [4]                            |
| [143] | 0.11           | NSFnets (Navier-Stokes flow nets): Physics-informed neural networks for the incompressible Navier-Stokes equations                                      | [1, 9]                         |

|       |      |                                                                                              |         |
|-------|------|----------------------------------------------------------------------------------------------|---------|
| [144] | 0.11 | Error estimates of residual minimization using neural networks for linear PDEs               | [2, 9]  |
| [145] | 0.11 | Solving PDEs by variational physics-informed neural networks: an a posteriori error analysis | [4, 9]  |
| [146] | 0.11 | Characterizing possible failure modes in physics-informed neural networks                    | [1, 11] |

## Adjacent Work

### Which papers cite the same foundational papers as relevant papers?

Use this table to discover related papers on adjacent topics, to gain a broader understanding of the field and help generate ideas for useful new research directions.

| Ref.  | Adjacency score | Title                                                                                                                | References These Foundational Papers |
|-------|-----------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| [154] | 1.74            | Building Trust in PINNs: Error Estimation through Finite Difference Methods                                          | [1, 4, 9, 11]                        |
| [88]  | 1.38            | PINNfluence: Interpreting PINNs through Influence Functions                                                          | [1, 9, 118, 146]                     |
| [21]  | 1.18            | Reliable Error Estimation for PINNs: Lower and Upper A Posteriori Bounds                                             | [1, 2, 22, 57, 118]                  |
| [17]  | 0.95            | Calibrated Physics-Informed Uncertainty Quantification                                                               | [71, 93, 113, 118, 139]              |
| [15]  | 0.86            | BEACONS: Bounded-Error, Algebraically-Composable Neural Solvers for Partial Differential Equations                   | [1, 4, 118]                          |
| [92]  | 0.61            | Uncertainty-Aware Diagnostics for Physics-Informed Machine Learning                                                  | [67, 118, 122, 141, 143, 146]        |
| [122] | 0.56            | Physics-informed machine learning                                                                                    | [67, 94, 102, 118, 123, 141]         |
| [71]  | 0.56            | Physics-Informed Neural Operator for Learning Partial Differential Equations                                         | [98, 118, 141, 143, 148]             |
| [67]  | 0.51            | Modeling the Dynamics of PDE Systems with Physics-Constrained Deep Auto-Regressive Networks                          | [94, 100, 118, 141, 148]             |
| [53]  | 0.50            | Uncertainty quantification in scientific machine learning: Methods, metrics, and comparisons                         | [94, 102, 118, 122, 141]             |
| [77]  | 0.49            | Data-Efficient Neural Operator Training via Physics-Based Active Learning                                            | [41, 82, 98, 102, 118]               |
| [155] | 0.49            | Man, Machine, and Mathematics                                                                                        | [9]                                  |
| [156] | 0.49            | Estimation of the input-to-state stability gain functions from finite-dimensional approximations                     | [9]                                  |
| [157] | 0.49            | Fredholm Neural Networks for inverse problems in elliptic PDEs                                                       | [9]                                  |
| [158] | 0.49            | Physics-Informed Neural Networks for the Structural Analysis and Monitoring of Railway Bridges: A Systematic Review  | [9]                                  |
| [159] | 0.49            | Advancing Generalization in PINNs Through Latent-Space Representations                                               | [9]                                  |
| [160] | 0.49            | Magnetic Field Analysis of Reactor Based on TL-PIDL and Time Domain PIDL                                             | [9]                                  |
| [161] | 0.49            | PINNs error estimates for nonlinear equations in $\beta$ -smooth Banach spaces                                       | [9]                                  |
| [162] | 0.49            | Port-Hamiltonian Dynamic Mode Decomposition                                                                          | [9]                                  |
| [163] | 0.48            | Physics-informed deep learning for elasto-plasticity: Strong-form, energy-minimization, and state-aware formulations | [71, 118, 122]                       |

## References

- [1] [Efficient Error Certification for Physics-Informed Neural Networks](#)  
Francisco Eiras, ..., and M. P. Kumar. Unknown journal, 2023. 9 citations.  
97% Topic Match  
Proposes a scalable, post-training certification method for PINN worst-case residuals.  
Extends NN verification (-CROWN) to bound nonlinear PDE residuals and produce domain-wide worst case error certificates for several PDEs.  
Relevant as oracle free, property based offline V&V (post hoc residual contracts); lacks diagnostic attribution, admissibility gating, numerical artifact disentanglement, or safety/assurance framing.
- [2] [Trustworthy AI in numerics: On verification algorithms for neural network-based PDE solvers](#)  
E. Haugen, ..., and Anders C. Hansen. ArXiv, 2025. 1 citations.  
93% Topic Match  
Proposes a posteriori verification algorithms certifying NN PDE solvers to accuracy.  
Computes rigorous  $L^p$  norm and derivative estimates plus PDE residual bounds to certify or reject networks without an oracle.  
Highly relevant: oracle-free, offline V&V emphasis; mathematical guarantees and rejection/accept decisions; focuses on verification (not training) and applies to PINNs/neural operators, but is theoretical/numerical-analysis rather than applied diagnostics or failure-attribution.
- [3] [Rigorous Error Certification for Neural PDE Solvers: From Empirical Residuals to Solution Guarantees](#)  
Amartya Mukherjee, ..., and Jun Liu. ArXiv, 2026. 0 citations.  
90% Topic Match  
Proposes a residual-to-error certification framework for neural PDE solvers.  
Proves deterministic/probabilistic bounds translating certified residual, boundary, and initial errors into computable solution-space error guarantees, using formal-verification tools (e.g., dReal) to certify uniform residuals.  
Relevant as an oracle-free, pre-deployment certification method for PINNs/ELMs, but it does not treat physics admissibility tests (conservation, symmetry, scaling) as independent V&V, nor provide diagnostics for failed-check attribution, admissibility gating, or system-level assurance framing.
- [4] [A posteriori certification for neural network approximations to PDEs](#)  
Lewin Ernst, ..., and K. Urban. ArXiv, 2025. 3 citations.  
85% Topic Match  
Proposes a posteriori, oracle-free certification via computable error bounds for NN PDE surrogates.  
Derives rigorous upper/lower bounds by estimating the dual norm of the NN residual using Riesz representations on simpler embedded/enveloping surrogate domains.  
Relevant: offline, pre-deployment, physics-informed V&V (error bounds in problem-induced norm) and architecture-agnostic; however focuses on norm-based error certification for linear elliptic/parabolic PDEs—not property-based admissibility checks (conservation, monotonicity, typed gating) nor diagnostic attribution of failed physics-tests.
- [5] [Certified and accurate computation of function space norms of deep neural networks](#)  
J. Gründler, ..., and Philipp C. Petersen. ArXiv, 2026. 0 citations.  
79% Topic Match  
Proposes a deterministic, oracle-free framework for certified computation of neural network function-space norms.  
Combines interval-arithmetic enclosures on axis-aligned boxes, adaptive marking/refinement, and quadrature aggregation to yield guaranteed local and global bounds for  $L^p$  and  $W^{k,p}$  norms (function values, Jacobians, Hessians).  
Relevant as an offline, post-training property-based V&V tool (non-statistical certificates), with convergence proofs and PINN-residual applications; does not focus on failure-attribution, typed admissibility gating, or system-level assurance arguments.
- [6] [A certified wavelet-based physics-informed neural network for the solution of parameterized partial differential equations](#)  
No author found. ArXiv, 2022. 6 citations.  
74% Topic Match  
Proposes a certified, wavelet-based PINN with a computable a posteriori error bound.  
Represents the PDE residual in an adaptive wavelet basis and uses weighted expansion coefficients as both loss and explicit upper-bound estimator for the dual residual norm.  
Relevant partly: it's offline, pre-deployment certification of PINNs for parameterized PDEs, but focuses on deriving numerical error bounds (not oracle free physics-property tests or diagnostic attribution of failed admissibility checks).
- [7] [Neural functional a posteriori error estimates](#)  
V. Fanaskov, ..., and I. Oseledets. ArXiv, 2024. 7 citations.  
69% Topic Match  
Proposes a training-loss method that yields computable a posteriori error majorants for neural PDE surrogates.  
Achieves this by training networks to predict auxiliary physical fields so classical functional a posteriori error-estimate theory produces cheap, rigorous upper-bound estimates after training.  
Relevant because it provides oracle-free, post hoc error bounds for neural operators/PINNs (elliptic PDEs), but it focuses on embedding estimates into loss/training and not on independent property based V&V, failure attribution, admissibility gating, or safety-case integration.
- [8] [Prediction error certification for PINNs: Theory, computation, and application to Stokes flow](#)  
Birgit Hillebrecht and B. Unger. ArXiv, 2025. 4 citations.  
66% Topic Match  
Proposes a computable semigroup-based a posteriori error certificate for PINNs.  
Modifies prior input-to-state-stability error bounds and gives numerical algorithms to approximate required stability constants, demonstrated on Stokes flow around a cylinder.  
Relevant as an offline, pre-deployment, oracle-free error-certification approach for PINN surrogates (rigorous bounds rather than oracle comparison), but focuses on stability-constant estimation and global error bounds—not property-based admissibility tests, diagnostic attribution of failed checks, or typed/gating frameworks for safety assurance.
- [9] [Rigorous a Posteriori Error Bounds for PDE-Defined PINNs](#)  
Birgit Hillebrecht and B. Unger. IEEE Transactions on Neural Networks and Learning Systems, 2022. 12 citations.  
62% Topic Match  
Proposes a computable a posteriori upper bound on PINN prediction error.  
Derives error estimators using only PDE-system a priori information, no ground-truth oracle, and demonstrates on transport, heat, NSE, Klein–Gordon.  
Relevant: focuses on offline, oracle-free certification of PINNs; primarily linear-PDE theory extension to PDEs, with limited discussion of diagnostic attribution, admissibility gating, or false-alarm control.
- [10] [Error Bounds for Physics-Informed Neural Networks in Fokker-Planck PDEs](#)  
Chun-Wei Kong, ..., and Morteza Lahijanian. Unknown journal, 2024. 4 citations.  
56% Topic Match  
Develops error bounds for PINN approximations of Fokker–Planck PDEs.  
Constructs a theoretical recursive error-PINN framework and a practical one PINN bound provable and computable without reference solutions.

Relevant for oracle-free, worst case error estimation of PINN surrogates, but does not present property-based admissibility tests, failure-attribution, typed gating, or safety/assurance case methodology.

[11] [Residual-based error bound for physics-informed neural networks](#)

Shuheng Liu, ..., and P. Protopapas. ArXiv, 2023. 11 citations.

50% Topic Match

Proposes a residual-based, oracle-free error-bounding method for PINN solutions of ODEs/PDEs.

Derives rigorous bounds from equation structure and the network residual ( $Du = f + r$ ), with algorithms trading tightness against runtime.

Relevant: it's offline, pre-deployment, oracle-free and physics-derived; however focuses on error bounds (not property-based admissibility tests or failure-attribution/gating) and assumes exact BC/IC enforcement and mainly linear/first-order PDEs.

[12] [A posteriori analysis of neural network approximations](#)

Thomas Führer and Sergio Rojas. ArXiv, 2025. 2 citations.

43% Topic Match

Proposes an a posteriori error-estimation framework for neural-network PDE approximations.

Decomposes solution error into a computable discrete-residual projector ( $\cdot$ ) plus a complementary FEM-style estimator ( $\mu$ ) with practical upper bound  $\hat{\mu}$  usable as offline diagnostics or as loss terms without an external oracle.

Relevant because it is oracle-free, yields computable property-like error checks (checks depend on variational formulation and quadrature), discusses quadrature/mesh effects and boundary-treatment, but focuses on elliptic PDEs and assumes well-posed variational form—diagnostic attribution of failures (surrogate vs. numerical/assumption violation) is suggested via refinement/ $\mu$  vs  $\cdot$ -behavior but not developed into a full admissibility-gating or safety-assurance framework.

[13] [First Order System Least Squares Neural Networks](#)

J. Opschoor, ..., and Christoph Schwab. ArXiv, 2024. 4 citations.

39% Topic Match

Proposes a FoSLS residual-based framework for neural PDE surrogates.

Uses L2 least-squares of a first-order system as a computable, physics-rooted loss and a posteriori error estimator with localized residuals.

Relevant for oracle-free offline diagnostics and adaptive NN refinement, but lacks property-test suites, failure-attribution, admissibility gating, false alarm control, or safety-assurance integration.

[14] [Correctness Verification of Neural Networks Approximating Differential Equations](#)

P. Ellinas, ..., and Spyros Chatzivasileiadis. ArXiv, 2024. 3 citations.

35% Topic Match

Proposes an offline correctness-verification framework for NNs solving differential equations.

Uses finite-difference NN derivatives, PDE-residual bounding, IVP error-propagation, and a parallel branching solver combining incomplete CROWN bounds with gradient attack domain rejection.

Relevant for oracle-free worst case error guarantees of PINN/NN surrogates, but does not develop property-based admissibility tests, typed gating, failure-attribution, discretization artifact disentanglement, or assurance-case integration.

[15] [BEACONS: Bounded-Error, Algebraically-Composable Neural Solvers for Partial Differential Equations](#)

Jonathan Gorard, ..., and J. Juno. ArXiv, 2026. 0 citations.

32% Topic Match

Proposes a framework (BEACONS) for formally verified neural PDE solvers with bounded worst case error.

Constructs machine checkable certificates using method-of-characteristics bounds, rigorous L extrapolation estimates, and algebraic composition of shallow nets to ensure stability, convergence, and conservation.

Highly relevant: focuses on offline, pre-deployment formal V&V (oracle free certificates), provides diagnostic guarantees (conservation/stability), and uses theorem proving/code generation — but appears centered on formal proof-based guarantees rather than empirical property test attribution or numerical vs assumption failure diagnosis.

[16] [A unified framework for the error analysis of physics-informed neural networks](#)

Marius Zeinhofer, ..., and K. Mardal. IMA Journal of Numerical Analysis, 2024. 11 citations.

29% Topic Match

Proves a priori and a posteriori error bounds for PINNs on linear PDEs.

Develops an abstract bilinear-form framework (Céa-type lemma adaptation), coercivity/continuity conditions, and quadrature/collocation error controls; includes numerical PINN examples.

Relevant mainly for theoretical verification of PINN convergence/error; does NOT provide oracle-free, offline property-based V&V, admissibility gating, failure-attribution, or assurance methodology for safety critical surrogates.

[17] [Calibrated Physics-Informed Uncertainty Quantification](#)

Vignesh Gopakumar, ..., and M. Deisenroth. ArXiv, 2025. 8 citations.

28% Topic Match

Proposes a model-agnostic, offline physics-informed conformal-prediction (CP) UQ framework for neural PDE surrogates.

Uses PDE residual (Physics Residual Error, PRE) as the CP nonconformity score, with convolutional layers-as-finite-difference stencils to compute residuals and produce calibrated marginal/joint coverage without labelled data.

Relevant as an oracle-free, post-hoc property-based V&V tool with proven coverage guarantees, but does not present a systematic diagnostic/attribution methodology (admissibility gating, typing, numerical-artifact disentanglement, or non statistical false alarm control).

[18] [Application of SciML-Adapted PCMM to Deep Neural Network Surrogate Model Used for Aerodynamic Coefficient Prediction](#)

Jared Kirsch, ..., and W. Rider. Journal of Verification, Validation and Uncertainty Quantification, 2026. 0 citations.

25% Topic Match

Applies a SciML-adapted Predictive Capability Maturity Model (PCMM) for surrogate credibility assessment.

Uses RANS-trained DNN for NACA 0012 aerodynamics and evaluates five credibility elements (data, domain, explainability, validation, UQ) including SHAP, experimental comparison, and quantified run-to-run/numerical uncertainties.

Relevant as an offline, pre-deployment credibility framework for CFD surrogates, but it relies on comparative validation against experiments and emphasizes interpretability/UQ rather than oracle free, property-based admissibility tests or detailed failure-attribution diagnostics.

[19] [Wholistic Credibility Assessment of Machine Learning-Based Surrogate Model in an Aerodynamics Application](#)

Jared Kirsch, ..., and Nima Fathi. ASME 2025 Verification, Validation, and Uncertainty Quantification Symposium, 2025. 0 citations.

22% Topic Match

Proposes a holistic credibility assessment framework for neural surrogates in aerodynamics.

Combines traditional computational-physics credibility elements with ML accuracy and uncertainty evaluation, demonstrated on subsonic turbulent airfoil flow.

Paper focuses on surrogate credibility broadly (design/optimization context); unclear if tests are oracle-free property-based or include failure-attribution, typed admissibility gating, or pre-deployment, non-oracle diagnostics — check full text for whether physics-derived property tests (conservation, invariances, numerical-discretization robustness) are used as independent V&V and whether interpretation/false-alarm handling is addressed.

[20] [Credibility Assessment of Machine Learning-Based Surrogate Model Predictions on NACA 0012 Airfoil Flow](#)

Jared Kirsch, ..., and Nima Fathi. ASME 2024 Verification, Validation, and Uncertainty Quantification Symposium, 2024. 0 citations.

20% Topic Match

Evaluates surrogate credibility for NACA 0012 lift/drag predictions.

Trains a deep neural-network surrogate for CFD coefficients across Re and AoA, then applies best-practice credibility and V&V analyses.

Likely compares surrogate vs. CFD oracle and discusses credibility evidence; does not clearly state oracle-free, property-based offline tests or diagnostic attribution—probably somewhat related but not precisely within the requested oracle-free, property-test methodology.

[21] [Reliable Error Estimation for PINNs: Lower and Upper A Posteriori Bounds](#)

Ismail G. Huseynov, ..., and A. Bashirov. ArXiv, 2026. 0 citations.

18% Topic Match

Proposes computable oracle-free a posteriori two-sided error certificates for PINNs.

Derives explicit lower bounds (new) under localized strong-monotonicity and upper bounds under one-sided Lipschitz, using only the NN, ODE residuals, and local constants.

Relevant as an offline, oracle-free V&V method for PINN surrogates (ODEs only); discusses extension challenges to PDEs, diagnostics (signed-residual probes), and use of certificates for post-training checks and regularization—but does not provide full attribution framework for failed checks, numerical-discretization disentanglement, or safety-case integration.

[22] [Certified machine learning: A posteriori error estimation for physics-informed neural networks](#)

Birgit Hillebrecht and B. Unger. 2022 International Joint Conference on Neural Networks (IJCNN), 2022. 26 citations.

18% Topic Match

Proposes an oracle-free a posteriori error estimator to certify PINN predictions (for ODEs).

Derives rigorous upper bounds on PINN prediction error, decomposes contributions, and trains a secondary "Error NN" for cheaper estimation.

Relevant as offline, pre-deployment, property-based V&V for PINNs but limited to ODEs (no PDE/CFD), discusses numerical vs model-error decomposition and practical limitations.

[23] [Physics-based software analysis for safety-critical embedded applications: work-in-progress](#)

Philipp Göttlich and H. Reuss. Proceedings of the International Conference on Embedded Software Companion, 2019. 1 citations.

17% Topic Match

Abstract: Physics-based software systems are a new trend in automotive feature and control implementation. They are used for modern applications in the context of automated driving and ADAS. Implementing physical models on critical embedded targets (for example electronic control systems) causes many safety related challenges, as they are far more complex than ordinary control algorithms. The code needs to be secured appropriately. This paper proposes a new approach to include physical properties in the V&V (verification & validation) and testing process in order to analyze functional code behavior of physical systems on targets automatically. For this, physical properties are extracted from...

[24] [Synthesizing Sound and Precise Abstract Transformers for Nonlinear Hyperbolic PDE Solvers](#)

Jacob Laurel, ..., and Jan Hückelheim. Proceedings of the ACM on Programming Languages, 2025. 1 citations.

15% Topic Match

Abstract: Partial Differential Equations (PDEs) play a ubiquitous role in scientific computing and engineering. While numerical methods make solving PDEs tractable, these numerical solvers encounter several issues, particularly for hyperbolic PDEs. These issues arise from multiple sources including the PDE's physical model, which can lead to effects like shock wave formation, and the PDE solver's inherent approximations, which can introduce spurious numerical artifacts. These issues can cause the solver's program execution to crash (due to overflow) or return results with unacceptable levels of inaccuracy (due to spurious oscillations or dissipation). Moreover, these challenges are compounded by the nonlinear nature of many...

[25] [Physics-Informed Decision Framework for Reuse of Reclaimed Steel Members Under Uncertainty](#)

Sina Sarfarazi, ..., and F. Fabbrocino. Metals, 2026. 5 citations.

15% Topic Match

Abstract: Structural steel reuse can gain large embodied-carbon savings, yet it is still not widely adopted since approval depends on the quality of the evidence, how uncertainty is handled, and if the design requirements are followed, not just on resistance. Reclaimed members frequently lack dependable documentation regarding material grade, loading history, boundary conditions, connection status, and degradation. For reuse decisions, conservative default assumptions protect safety but frequently eliminate qualified reuse options. This research examines data-driven and physics-informed computational methods from a decision-making standpoint, contending that their significance resides in facilitating an auditable approval process, not in supplanting deterministic verification. We differentiate...

[26] [Surrogate Neural Networks Local Stability for Aircraft Predictive Maintenance](#)

Mélanie Ducoffe, ..., and Eugene O'Higgins. ArXiv, 2024. 4 citations.

15% Topic Match

Abstract: Surrogate Neural Networks are nowadays routinely used in industry as substitutes for computationally demanding engineering simulations (e.g., in structural analysis). They allow to generate faster predictions and thus analyses in industrial applications e.g., during a product design, testing or monitoring phases. Due to their performance and time-efficiency, these surrogate models are now being developed for use in safety-critical applications. Neural network verification and in particular the assessment of their robustness (e.g., to perturbations) is the next critical step to allow their inclusion in real-life applications and certification. We assess the applicability and scalability of empirical and formal methods in the...

[27] [HOLISTIC CREDIBILITY ASSESSMENT OF A MACHINE LEARNING-BASED SURROGATE MODEL IN AN AERODYNAMICS APPLICATION](#)

Jared Kirsch, ..., and Nima Fathi. Journal Not Provided, 2025. 0 citations.

14% Topic Match

No summary or abstract available

[28] [Integrating Physical Systems in the Static Analysis of Embedded Control Software](#)

P. Cousot. Unknown journal, 2005. 26 citations.

14% Topic Match

No summary or abstract available

[29] [A Framework for Strategic Discovery of Credible Neural Network Surrogate Models under Uncertainty](#)

P. Singh, ..., and D. Faghihi. ArXiv, 2024. 16 citations.

14% Topic Match

Abstract: The widespread integration of deep neural networks in developing data-driven surrogate models for high-fidelity simulations of complex physical systems highlights the critical necessity for robust uncertainty quantification techniques and credibility assessment methodologies, ensuring the reliable deployment of surrogate models in consequential decision-making. This study presents the Occam Plausibility Algorithm for surrogate models (OPAL-surrogate), providing a systematic framework to uncover predictive neural network-based surrogate models within the large space of potential models, including various neural network classes and choices of architecture and hyperparameters. The framework is grounded in hierarchical Bayesian inferences and employs model validation tests to evaluate the credibility and...

[30] [Exact and approximate error bounds for physics-informed neural networks](#)

Augusto T. Chantada, ..., and C. Sc'occola. ArXiv, 2024. 3 citations.

13% Topic Match

Abstract: The use of neural networks to solve differential equations, as an alternative to traditional numerical solvers, has increased recently. However, error bounds for the obtained solutions have only been developed for certain equations. In this work, we report important progress in calculating error bounds of physics-informed neural networks (PINNs) solutions of nonlinear first-order ODEs. We give a general expression that describes the error of the solution that the PINN-based method provides for a nonlinear first-order ODE. In addition, we propose a technique to calculate an approximate bound for the general case and an exact bound for a particular case. The...

[31] [Active learning surrogates in system reliability analysis: a review](#)

Hong Zhang and Lukai Song. International Journal of Structural Integrity, 2025. 3 citations.

12% Topic Match

Abstract: This review aims to comprehensively summarize recent progress in surrogate modeling for reliability analysis of complex mechanical systems, particularly aeroengine mechanisms operating under harsh, multi-physics conditions. It focuses on how active learning strategies enhance modeling efficiency and accuracy, addressing challenges such as small failure probabilities, multidisciplinary coupling, high-dimensional inputs and time-varying uncertainties. The study reviews regression-based surrogates (e.g. Gaussian process regression, artificial neural networks, random forest regression) and interpolation-based surrogates (e.g. radial

basis function, Kriging), outlining their strengths and limitations. It examines active learning methods – such as expected improvement (EI), upper confidence bound (UCB) and reliability-based expected improvement...

[32] [Agentic Physical AI toward a Domain-Specific Foundation Model for Energy Systems: A Case Study on Nuclear Reactor Control](#)

Yoonpyo Lee, ..., and S. Alam. ArXiv, 2025. 1 citations.

12% Topic Match

Abstract: The prevailing paradigm in AI for physical systems: scaling general-purpose foundation models toward universal multimodal reasoning, confronts a barrier at the control interface. Frontier vision-language models achieve only 50-53% accuracy on basic quantitative physics tasks, behaving as approximate guessers that preserve semantic plausibility while violating physical constraints. Safety-critical control demands outcome-space guarantees over executed actions, not parameter-space imitation. Here we present a pathway toward domain-specific foundation models through compact language models operating as Agentic Physical AI: policy optimization driven by physics-based simulator validation rather than perceptual inference. We train a 360M-parameter model on synthetic nuclear reactor scenarios scaled from  $10^3$ ...

[33] [Evaluating Error Bound for Physics-Informed Neural Networks on Linear Dynamical Systems](#)

Shuheng Liu, ..., and P. Protopapas. ArXiv, 2022. 6 citations.

11% Topic Match

Abstract: There have been extensive studies on solving differential equations using physics-informed neural networks. While this method has proven advantageous in many cases, a major criticism lies in its lack of analytical error bounds. Therefore, it is less credible than its traditional counterparts, such as the finite difference method. This paper shows that one can mathematically derive explicit error bounds for physics-informed neural networks trained on a class of linear systems of differential equations. More importantly, evaluating such error bounds only requires evaluating the differential equation residual infinity norm over the domain of interest. Our work shows a link between network...

[34] [Automating Physics-Based Reasoning for SysML Model Validation](#)

C. Chambers, ..., and S. Bhattacharyya. 2025 IEEE International systems Conference (SysCon), 2025. 1 citations.

11% Topic Match

Abstract: System and software design benefits greatly from formal modeling, allowing for automated analysis and verification early in the design phase. Current methods excel at checking information flow and component interactions, ensuring consistency, and identifying dependencies within Systems Modeling Language (SysML) models. However, these approaches often lack the capability to perform physics-based reasoning about a system's behavior represented in SysML models, particularly in the electromechanical domain. This significant gap critically hinders the ability to automatically and effectively verify the correctness and consistency of the model's behavior against well-established underlying physical principles. Therefore, this paper presents an approach that leverages existing research...

[35] [Functionality and Safety Weaknesses in Integration of Physics-Based Models on Critical Embedded Systems](#)

Philipp Göttlich and H. Reuss. 2019 IEEE International Symposium on Software Reliability Engineering Workshops (ISSREW), 2019. 0 citations.

10% Topic Match

Abstract: Embedded automotive software is currently showing trends towards model predictive control (MPC), virtual sensors or model-based diagnosis, mainly used in advanced driver assistance systems (ADAS) and automated driving. Such applications use physical models in the control algorithms. The integration of physical models is a risky task, since weaknesses, such as the need for floating-point arithmetic and discretization or model properties, such as discontinuities and nonlinearities, quickly bring a project to a standstill or establish errors in the final product. The use of known verification and validation methods is often not possible or offers false safety guarantees. This article is intended...

[36] [Shock with Confidence: Formal Proofs of Correctness for Hyperbolic Partial Differential Equation Solvers](#)

Jonathan Gorard and Ammar Hakim. ArXiv, 2025. 3 citations.

10% Topic Match

Abstract: First-order systems of hyperbolic partial differential equations (PDEs) occur ubiquitously throughout computational physics, commonly used in simulations of fluid turbulence, shock waves, electromagnetic interactions, and even general relativistic phenomena. Such equations are often challenging to solve numerically in the non-linear case, due to their tendency to form discontinuities even for smooth initial data, which can cause numerical algorithms to become unstable, violate conservation laws, or converge to physically incorrect solutions. In this paper, we introduce a new formal verification pipeline for such algorithms in Racket, which allows a user to construct a bespoke hyperbolic PDE solver for a specified equation...

[37] [Model-Based Software Engineering: Physics Coupled with Software for Test Case Reuse Across the V Cycle](#)

Jordan McBain. SAE Technical Paper Series, 2025. 0 citations.

10% Topic Match

Abstract: The development of cyber-physical systems necessarily involves the expertise of an interdisciplinary team – not all of whom have deep embedded software knowledge. Graphical software development environments alleviate many of these challenges but in turn create concerns for their appropriateness in a rigorous software initiative. Their tool suites further enable the creation of physics models which can be coupled in the loop with the corresponding software component's control law in an integrated test environment. Such a methodology addresses many of the challenges that arise in trying to create suitable test cases for physics-based problems. If the test developer ensures that...

[38] [A Unified Framework for the Error Analysis of Physics-Informed Neural Networks](#)

Marius Zeinhofer, ..., and K. Mardal. ArXiv, 2023. 24 citations.

9% Topic Match

Abstract: We prove a priori and a posteriori error estimates for physics-informed neural networks (PINNs) for linear PDEs. We analyze elliptic equations in primal and mixed form, elasticity, parabolic, hyperbolic and Stokes equations; and a PDE constrained optimization problem. For the analysis, we propose an abstract framework in the common language of bilinear forms, and we show that coercivity and continuity lead to error estimates. The obtained estimates are sharp and reveal that the  $L^2$  penalty approach for initial and boundary conditions in the PINN formulation weakens the norm of the error decay. Finally, utilizing recent advances in PINN optimization, we...

[39] [Error analysis for physics-informed neural networks \(PINNs\) approximating Kolmogorov PDEs](#)

Tim De Ryck and Siddhartha Mishra. Advances in Computational Mathematics, 2021. 138 citations.

8% Topic Match

Abstract: Physics-informed neural networks approximate solutions of PDEs by minimizing pointwise residuals. We derive rigorous bounds on the error, incurred by PINNs in approximating the solutions of a large class of linear parabolic PDEs, namely Kolmogorov equations that include the heat equation and Black-Scholes equation of option pricing, as examples. We construct neural networks, whose PINN residual (generalization error) can be made as small as desired. We also prove that the total  $L^2$ -error can be bounded by the generalization error, which in turn is bounded in terms of the training error, provided that a sufficient number of randomly chosen training (collocation)...

[40] [Physics-Informed Mixed-Criticality Scheduling for F1Tenth Cars with Preemptable ROS 2 Executors](#)

Kurt M. Wilson, ..., and Zhishan Guo. 2025 IEEE 31st Real-Time and Embedded Technology and Applications Symposium (RTAS), 2025. 8 citations.

8% Topic Match

Abstract: Autonomous systems are increasingly used in safety-critical domains, including industrial automation, autonomous vehicles, and the industrial Internet of Things. Verifying both the functional and temporal correctness of these systems is essential to ensure safety before deployment. However, end-to-end verification is challenging due to the interaction of continuous-time physical processes with discrete-time computational systems. Existing formal methods often assume simplified or static computational models, while traditional real-time systems focus on meeting timing constraints without explicitly linking them to physical safety. We address this gap by proposing a physics-informed mixed-criticality (MC) verification framework for cyber-physical systems, which allows the integration of computational...

[41] [PDEBENCH: An Extensive Benchmark for Scientific Machine Learning](#)

M. Takamoto, ..., and Mathias Niepert. ArXiv, 2022. 471 citations.

8% Topic Match

Abstract: Machine learning-based modeling of physical systems has experienced increased interest in recent years. Despite some impressive progress, there is still a lack of benchmarks for Scientific ML that are easy to use but still challenging and representative of a wide range of problems. We introduce PDEBench, a benchmark suite of time-dependent simulation tasks based on Partial Differential Equations (PDEs). PDEBench comprises both code and data to benchmark

the performance of novel machine learning models against both classical numerical simulations and machine learning baselines. Our proposed set of benchmark problems contribute the following unique features: (1) A much wider range of...

[42] [Non-asymptotic stability and consistency guarantees for physics-informed neural networks via coercive operator analysis](#)

Ronald Katende. Commun. Nonlinear Sci. Numer. Simul., 2025. 0 citations.

8% Topic Match

Abstract: We present a unified theoretical framework for analyzing the stability and consistency of Physics-Informed Neural Networks (PINNs), grounded in operator coercivity, variational formulations, and non-asymptotic perturbation theory. PINNs approximate solutions to partial differential equations (PDEs) by minimizing residual losses over sampled collocation and boundary points. We formalize both operator-level and variational notions of consistency, proving that residual minimization in Sobolev norms leads to convergence in energy and uniform norms under mild regularity. Deterministic stability bounds quantify how bounded perturbations to the network outputs propagate through the full composite loss, while probabilistic concentration results via McDiarmid's inequality yield sample complexity guarantees...

[43] [Steady-State Transonic Flowfield Prediction via Deep-Learning Framework](#)

Gabriele Immordino, ..., and Marcello Righi. AIAA Journal, 2024. 16 citations.

7% Topic Match

Abstract: This paper focuses on the development of a deep-learning framework for predicting distributed quantities around aircraft flying in the transonic regime. These quantities play a crucial role in determining aerodynamic loads and conducting aeroelastic analysis. Angle of attack and Mach number are chosen as the two independent parameters for the reduced-order models. A comparative assessment is conducted between the proposed deep-learning framework and the proper orthogonal decomposition approach to identify the strengths and weaknesses of each method. The accuracy of the data-driven machine-learning method in modeling steady-state transonic aerodynamics is assessed against three benchmark cases of three-dimensional test cases: Benchmark...

[44] [Validation-Gated Multi-Agent Governance for Online Adaptation of Thermal-Hydraulic Surrogate Models under Operating-Regime Shift](#)

Doyeong Lim, ..., and In-Cheol Bang. ArXiv, 2026. 0 citations.

6% Topic Match

Abstract: Artificial-intelligence surrogates can support second-by-second thermal-hydraulic forecasting, but models selected and frozen offline may become condition-locked once deployed outside their pretraining envelope. This study develops a guarded continual-adaptation framework for experimental thermal-hydraulic loop data in which role-separated agents - Monitor, Diagnosis, Adaptation, Safety-Auditor, and Orchestrator - diagnose error signatures, prioritize candidate model families, and review promotions, while deterministic champion-challenger gates and background shadow learning retain final authority over model replacement. Seven surrogate families were screened by blocked three-fold cross-validation, and a temporal Fourier neural operator was selected as the initial champion for 60-s-history-to-10-s-trajectory forecasting on two held-out transients, with three...

[45] [Estimates on the generalization error of physics-informed neural networks for approximating a class of inverse problems for PDEs](#)

Siddhartha Mishra and R. Molinaro. IMA Journal of Numerical Analysis, 2020. 350 citations.

6% Topic Match

Abstract: Physics-informed neural networks (PINNs) have recently been very successfully applied for efficiently approximating inverse problems for partial differential equations (PDEs). We focus on a particular class of inverse problems, the so-called data assimilation or unique continuation problems, and prove rigorous estimates on the generalization error of PINNs approximating them. An abstract framework is presented and conditional stability estimates for the underlying inverse problem are employed to derive the estimate on the PINN generalization error, providing rigorous justification for the use of PINNs in this context. The abstract framework is illustrated with examples of four prototypical linear PDEs. Numerical experiments,...

[46] [Learning Data-Efficient and Generalizable Neural Operators via Fundamental Physics Knowledge](#)

Siyang Ma, ..., and V. Ganesh. ArXiv, 2026. 3 citations.

6% Topic Match

Abstract: Recent advances in scientific machine learning (SciML) have enabled neural operators (NOs) to serve as powerful surrogates for modeling the dynamic evolution of physical systems governed by partial differential equations (PDEs). While existing approaches focus primarily on learning simulations from the target PDE, they often overlook more fundamental physical principles underlying these equations. Inspired by how numerical solvers are compatible with simulations of different settings of PDEs, we propose a multiphysics training framework that jointly learns from both the original PDEs and their simplified basic forms. Our framework enhances data efficiency, reduces predictive errors, and improves out-of-distribution (OOD) generalization, particularly...

[47] [Estimates on the generalization error of Physics Informed Neural Networks \(PINNs\) for approximating PDEs](#)

Siddhartha Mishra and R. Molinaro. ArXiv, 2020. 234 citations.

6% Topic Match

Abstract: Physics informed neural networks (PINNs) have recently been widely used for robust and accurate approximation of PDEs. We provide rigorous upper bounds on the generalization error of PINNs approximating solutions of the forward problem for PDEs. An abstract formalism is introduced and stability properties of the underlying PDE are leveraged to derive an estimate for the generalization error in terms of the training error and number of training samples. This abstract framework is illustrated with several examples of nonlinear PDEs. Numerical experiments, validating the proposed theory, are also presented.

[48] [Exactness Matters for Physical Rule Enforcement](#)

Bum Jun Kim. ArXiv, 2026. 0 citations.

5% Topic Match

Abstract: Autoregressive scientific forecasters often enforce physical or structural constraints by repairing each predicted state before feeding it back into the model. However, it remains unclear when stronger physical rule enforcement becomes reliable and when it becomes a source of distribution shift. We study this question through operator exactness, meaning whether the repair map is the identity on the target manifold and is aligned with the target geometry. We compare raw forecasting, post hoc repair, and in-loop repair across periodic incompressible Navier-Stokes, non-periodic CFDBench flows, and a hierarchical-forecasting support task. In the exact periodic regime, Fourier projection substantially improves rollout accuracy...

[49] [Assessment of Multifidelity Surrogate Approaches for Expedient Loads Prediction in High-Speed Flows](#)

Tyler E. Korenyi-Both, ..., and Daniel A. Reasor. Journal of Spacecraft and Rockets, 2024. 2 citations.

5% Topic Match

Abstract: Fast and accurate predictions about the flowfield surrounding a hypersonic flight vehicle are necessary for both early design phases and autonomous vehicle control. Data-driven model reduction presents an opportunity to harness the accuracy of high-fidelity techniques for expedient online application. Surrogate modeling is one approach that seeks to emulate the response of a system with evaluation times much faster than the data source. Multifidelity surrogate modeling seeks to reduce the amount of high-fidelity data needed for a sufficiently accurate online model. The objective of this paper is to assess the tradeoffs between a kriging-based multifidelity model and a neural-network-based multifidelity...

[50] [Guaranteed Prediction Sets for Functional Surrogate Models](#)

Ander Gray, ..., and Sébastien Destercke. Unknown journal, 2025. 4 citations.

5% Topic Match

Abstract: We propose a method for obtaining statistically guaranteed prediction sets for functional machine learning methods: surrogate models which map between function spaces, motivated by the need to build reliable PDE emulators. The method constructs nested prediction sets on a low-dimensional representation (an SVD) of the surrogate model's error, and then maps these sets to the prediction space using set-propagation techniques. This results in prediction sets for functional surrogate models with conformal prediction coverage guarantees. We use zonotopes as basis of the set construction, which allow an exact linear propagation and are closed under Cartesian products, making them well-suited to this...

[51] [Robust error estimates of PINN in one-dimensional boundary value problems for linear elliptic equations](#)

Jihahm Yoo and Haesung Lee. ArXiv, 2024. 5 citations.

5% Topic Match

Abstract: In this paper, we study physics-informed neural networks (PINN) to approximate solutions to one-dimensional boundary value problems for linear elliptic equations and establish robust error estimates of PINN regardless of the quantities of the coefficients. In particular, we rigorously demonstrate the existence and uniqueness of solutions using the Sobolev space theory based on a variational approach. Deriving  $L^2$ -contraction estimates, we show that the error, defined

as the mean square of the differences between the true solution and our trial function at the sample points, is dominated by the training loss. Furthermore, we show that as the quantities of...

[52] [A Mathematical Guide to Operator Learning](#)

Nicolas Boule and Alex Townsend. ArXiv, 2023. 101 citations.

5% Topic Match

Abstract: Operator learning aims to discover properties of an underlying dynamical system or partial differential equation (PDE) from data. Here, we present a step-by-step guide to operator learning. We explain the types of problems and PDEs amenable to operator learning, discuss various neural network architectures, and explain how to employ numerical PDE solvers effectively. We also give advice on how to create and manage training data and conduct optimization. We offer intuition behind the various neural network architectures employed in operator learning by motivating them from the point-of-view of numerical linear algebra.

[53] [Uncertainty quantification in scientific machine learning: Methods, metrics, and comparisons](#)

Apostolos F. Psaros, ..., and G. Karniadakis. J. Comput. Phys., 2022. 444 citations.

4% Topic Match

Abstract: Neural networks (NNs) are currently changing the computational paradigm on how to combine data with mathematical laws in physics and engineering in a profound way, tackling challenging inverse and ill-posed problems not solvable with traditional methods. However, quantifying errors and uncertainties in NN-based inference is more complicated than in traditional methods. This is because in addition to aleatoric uncertainty associated with noisy data, there is also uncertainty due to limited data, but also due to NN hyperparameters, overparametrization, optimization and sampling errors as well as model misspecification. Although there are some recent works on uncertainty quantification (UQ) in NNs, there...

[54] [On Constructing Efficient UAV Aerodynamic Surrogate Models for Digital Twins](#)

D. Aláez, ..., and J. J. Astrain. IEEE Transactions on Industrial Informatics, 2024. 4 citations.

4% Topic Match

Abstract: Aerodynamic modeling and optimization for unmanned aerial vehicles (UAVs) are complex and computationally intensive tasks. Surrogate models have emerged as a powerful tool for increasing efficiency in the aircraft design and optimization process. We review and evaluate some modeling techniques, such as artificial neural networks and support vector regression, showing that Gaussian process regression generally provides a well-performing solution to this type of problem. We propose an active learning algorithm based on the relevance factor, that combines bias estimated from nearest-neighbor Euclidean distance and variance, to achieve higher accuracy with fewer computational fluid dynamics (CFD) simulations. The obtained performance is...

[55] [Physics-Informed AI in Python: Neural Operators with Verified Error Bounds](#)

Murali Krishna Pasupuleti. International Journal of Academic and Industrial Research Innovations(IJAIRI), 2023. 0 citations.

4% Topic Match

Abstract: Abstract: This study addresses the lack of trustworthy uncertainty quantification in physics-informed neural operators used to approximate solution operators of partial differential equations. A Python framework is proposed in which Fourier- and DeepONet-style backbones are trained with physics residuals, energy-consistency penalties, and boundary/initial condition constraints, while a verification layer delivers instance-wise error guarantees. The method integrates global Lipschitz constant estimation, interval bound propagation over input function spaces, and distribution-free conformal calibration to produce certified prediction sets. Experiments on synthetic PDE benchmarks (elliptic, parabolic, and advection-diffusion systems) evaluate accuracy and coverage using relative L2 error, energy norm error, certified upper bounds,...

[56] [Unified generalization analysis for physics informed neural networks](#)

Yuka Hashimoto and Tomoharu Iwata. ArXiv, 2026. 0 citations.

4% Topic Match

Abstract: Physics-Informed Neural Networks (PINNs) and their variational counterparts (VPINNs) are neural networks that incorporate physical laws, making them useful for scientific problems. Existing generalization analyses for PINNs and VPINNs remain limited, often requiring restrictive assumptions such as stability conditions or linear ellipticity. In this paper, we derive generalization bounds for neural networks that involve differentiation with respect to input variables, covering PINNs and VPINNs under a unified framework. We apply Taylor expansion to represent nonlinear differential operators as linear operators on a high-dimensional space, enabling the use of Koopman-based analysis and showing that high-rank networks can generalize well even in...

[57] [Generic bounds on the approximation error for physics-informed \(and\) operator learning](#)

T. D. Ryck and Siddhartha Mishra. ArXiv, 2022. 88 citations.

4% Topic Match

Abstract: We propose a very general framework for deriving rigorous bounds on the approximation error for physics-informed neural networks (PINNs) and operator learning architectures such as DeepONets and FNOs as well as for physics-informed operator learning. These bounds guarantee that PINNs and (physics-informed) DeepONets or FNOs will efficiently approximate the underlying solution or solution operator of generic partial differential equations (PDEs). Our framework utilizes existing neural network approximation results to obtain bounds on more involved learning architectures for PDEs. We illustrate the general framework by deriving the first rigorous bounds on the approximation error of physics-informed operator learning and by showing...

[58] [Learning the solution operator of parametric partial differential equations with physics-informed DeepONets](#)

Sifan Wang, ..., and P. Perdikaris. Science Advances, 2021. 1105 citations.

3% Topic Match

Abstract: Enabling the rapid emulation of parametric differential equations with physics-informed deep operator networks.

[59] [Error Estimates and Generalized Trial Constructions for Solving ODEs Using Physics-Informed Neural Networks](#)

A. Babni, ..., and J. A. Rodrigues. Mathematical and Computational Applications, 2025. 2 citations.

3% Topic Match

Abstract: In this paper, we address the challenge of solving differential equations using physics-informed neural networks (PINNs), an innovative approach that integrates known physical laws into neural network training. The PINN approach involves three main steps: constructing a neural-network-based solution ansatz, defining a suitable loss function, and minimizing this loss via gradient-based optimization. We review two primary PINN formulations: the standard PINN I and an enhanced PINN II. The latter explicitly incorporates initial, final, or boundary conditions. Focusing on first-order differential equations, PINN II methods typically express the approximate solution as  $u^*(x, \cdot) = \mathcal{P}(x) + Q(x)N(x, \cdot)$ , where  $N(x, \cdot)$  is the neural network output with parameters...

[60] [Multi-fidelity Deep Learning-based methodology for epistemic uncertainty quantification of turbulence models](#)

Minghan Chu and Weicheng Qian. ArXiv, 2023. 2 citations.

3% Topic Match

Abstract: Computational Fluid Dynamics (CFD) simulations using turbulence models are commonly used in engineering design. Of the different turbulence modeling approaches that are available, eddy viscosity based models are the most common for their computational economy. Eddy viscosity based models utilize many simplifications for this economy such as the gradient diffusion and the isotropic eddy viscosity hypotheses. These simplifications limit the degree to which eddy viscosity models can replicate turbulence physics and lead to model form uncertainty. The Eigenspace Perturbation Method (EPM) has been developed for purely physics based estimates of this model form uncertainty in turbulence model predictions. Due to...

[61] [NeuralUQ: A comprehensive library for uncertainty quantification in neural differential equations and operators](#)

Zongren Zou, ..., and G. Karniadakis. ArXiv, 2022. 55 citations.

3% Topic Match

Abstract: Uncertainty quantification (UQ) in machine learning is currently drawing increasing research interest, driven by the rapid deployment of deep neural networks across different fields, such as computer vision, natural language processing, and the need for reliable tools in risk-sensitive applications. Recently, various machine learning models have also been developed to tackle problems in the field of scientific computing with applications to computational science and engineering (CSE). Physics-informed neural networks and deep operator networks are two such models for solving partial differential equations and learning operator mappings, respectively. In this regard, a comprehensive study of UQ methods tailored specifically for scientific...

[62] [Towards Size-Independent Generalization Bounds for Deep Operator Nets](#)

Pulkit Gopalani, ..., and Anirbit Mukherjee. Trans. Mach. Learn. Res., 2022. 7 citations.

3% Topic Match

Abstract: In recent times machine learning methods have made significant advances in becoming a useful tool for analyzing physical systems. A particularly active area in this theme has been "physics-informed machine learning" which focuses on using neural nets for numerically solving differential equations. In this work, we aim to advance the theory of measuring out-of-sample error while training DeepONets - which is among the most versatile ways to solve P.D.E systems in one-shot. Firstly, for a class of DeepONets, we prove a bound on their Rademacher complexity which does not explicitly scale with the width of the nets involved. Secondly, we use this...

[63] [Uncertainty quantification of surrogate models using conformal prediction](#)

Vignesh Gopakumar, ..., and M. Deisenroth. Machine Learning: Science and Technology, 2024. 13 citations.

3% Topic Match

Abstract: Data-driven surrogate models offer fast, inexpensive approximations to complex numerical and experimental systems but typically lack uncertainty quantification, limiting their reliability in safety-critical applications. While Bayesian methods provide uncertainty estimates, they offer no statistical guarantees and struggle with high-dimensional spatio-temporal problems due to computational costs and dependence on prior specification. We present a conformal prediction (CP) framework that provides statistically guaranteed marginal coverage for surrogate models in a model-agnostic manner with near-zero computational cost. Our approach handles high-dimensional spatio-temporal outputs by performing cell-wise calibration while preserving the tensorial structure of predictions. Through extensive empirical evaluation across diverse applications-including partial differential...

[64] [Investigating the ability of PINNs to solve Burgers' PDE near finite-time blowup](#)

Dibyakanti Kumar and Anirbit Mukherjee. Machine Learning: Science and Technology, 2023. 4 citations.

3% Topic Match

Abstract: Physics Informed Neural Networks (PINNs) have been achieving ever newer feats of solving complicated Partial Differential Equations (PDEs) numerically while offering an attractive trade-off between accuracy and speed of inference. A particularly challenging aspect of PDEs is that there exist simple PDEs which can evolve into singular solutions in finite time starting from smooth initial conditions. In recent times some striking experiments have suggested that PINNs might be good at even detecting such finite-time blow-ups. In this work, we embark on a program to investigate this stability of PINNs from a rigorous theoretical viewpoint. Firstly, we derive error bounds for...

[65] [Surrogate modeling of multi-dimensional premixed and non-premixed combustion using pseudo-time stepping physics-informed neural networks](#)

Zhen Cao, ..., and Jianren Fan. Physics of Fluids, 2024. 11 citations.

3% Topic Match

Abstract: Physics-informed neural networks (PINNs) have emerged as a promising alternative to conventional computational fluid dynamics (CFD) approaches for solving and modeling multi-dimensional flow fields. They offer instant inference speed and cost-effectiveness without the need for training datasets. However, compared to common data-driven methods, purely learning the physical constraints of partial differential equations and boundary conditions is much more challenging and prone to convergence issues leading to incorrect local optima. This training robustness issue significantly increases the difficulty of fine-tuning PINNs and limits their widespread adoption. In this work, we present improvements to the prior field-resolving surrogate modeling framework for combustion...

[66] [Active learning of deep surrogates for PDEs: application to metasurface design](#)

R. Pestourie, ..., and Steven G. Johnson. npj Computational Materials, 2020. 103 citations.

3% Topic Match

Abstract: Surrogate models for partial differential equations are widely used in the design of metamaterials to rapidly evaluate the behavior of composable components. However, the training cost of accurate surrogates by machine learning can rapidly increase with the number of variables. For photonic-device models, we find that this training becomes especially challenging as design regions grow larger than the optical wavelength. We present an active-learning algorithm that reduces the number of simulations required by more than an order of magnitude for an NN surrogate model of optical-surface components compared to uniform random samples. Results show that the surrogate evaluation is over...

[67] [Modeling the Dynamics of PDE Systems with Physics-Constrained Deep Auto-Regressive Networks](#)

N. Geneva and N. Zabarar. ArXiv, 2019. 343 citations.

2% Topic Match

Abstract: Abstract In recent years, deep learning has proven to be a viable methodology for surrogate modeling and uncertainty quantification for a vast number of physical systems. However, in their traditional form, such models can require a large amount of training data. This is of particular importance for various engineering and scientific applications where data may be extremely expensive to obtain. To overcome this shortcoming, physics-constrained deep learning provides a promising methodology as it only utilizes the governing equations. In this work, we propose a novel auto-regressive dense encoder-decoder convolutional neural network to solve and model non-linear dynamical systems without training...

[68] [A Priori Analysis of Stable Neural Network Solutions to Numerical PDEs](#)

Q. Hong, ..., and Jinchao Xu. ArXiv, 2021. 30 citations.

2% Topic Match

Abstract: Methods for solving PDEs using neural networks have recently become a very important topic. We provide an a priori error analysis for such methods which is based on the  $\| \cdot \|_{1, \infty}$ -norm of the solution. We show that the resulting constrained optimization problem can be efficiently solved using a greedy algorithm, which replaces stochastic gradient descent. Following this, we show that the error arising from discretizing the energy integrals is bounded both in the deterministic case, i.e. when using numerical quadrature, and also in the stochastic case, i.e. when sampling points to approximate the integrals. In the later case, we use a...

[69] [Multi-fidelity aerodynamic data fusion by autoencoder transfer learning](#)

Javier Nieto-Centenero, ..., and Rodrigo Castellanos. ArXiv, 2025. 0 citations.

2% Topic Match

Abstract: Accurate aerodynamic prediction often relies on high-fidelity simulations; however, their prohibitive computational costs severely limit their applicability in data-driven modeling. This limitation motivates the development of multi-fidelity strategies that leverage inexpensive low-fidelity information without compromising accuracy. Addressing this challenge, this work presents a multi-fidelity deep learning framework that combines autoencoder-based transfer learning with a newly developed Multi-Split Conformal Prediction (MSCP) strategy to achieve uncertainty-aware aerodynamic data fusion under extreme data scarcity. The methodology leverages abundant Low-Fidelity (LF) data to learn a compact latent physics representation, which acts as a frozen knowledge base for a decoder that is subsequently fine-tuned using...

[70] [Learning-Enhanced Finite Volume Methods for Nonlinear Convection-Diffusion Problems](#)

M. Al-asadi. Dijlah Journal of Engineering Sciences ISSN: 3078-9664, e-ISSN: 3078-9656, 2026. 0 citations.

2% Topic Match

Abstract: Background: The non-linear convection-diffusion equation is one of the cornerstones of computational fluid dynamics and applied mathematics, particularly in applications such as environmental modeling, heat transfer or reactive flows. The finite volume method (FVM) is known for its good conservation property and robustness, but usually has difficulty in accurately simulating the sharp gradients and discontinuities without high computational cost. PINNs also provide a mesh-free data-driven technique however it may suffer from the slow convergence rate and poor generalization. To address these shortcomings, in this manuscript we introduce Learning-Enhanced Finite Volume Methods (LE-FVM), a new class of approach that bridges the...

[71] [Physics-Informed Neural Operator for Learning Partial Differential Equations](#)

Zong-Yi Li, ..., and Anima Anandkumar. ACM/IMS Journal of Data Science, 2021. 831 citations.

2% Topic Match

Abstract: In this article, we propose physics-informed neural operators (PINO) that combine training data and physics constraints to learn the solution operator of a given family of parametric Partial Differential Equations (PDE). PINO is the first hybrid approach incorporating data and PDE constraints at different resolutions to learn the operator. Specifically, in PINO, we combine coarse-resolution training data with PDE constraints imposed at a higher resolution. The resulting PINO model can accurately approximate the ground-truth solution operator for many popular PDE families and shows no degradation in accuracy even under zero-shot super-resolution, that is, being able to predict beyond the resolution...

[72] [Loss Landscape Engineering via Data Regulation on PINNs](#)

Vignesh Gopakumar, ..., and D. Samaddar. ArXiv, 2022. 30 citations.

2% Topic Match

Abstract: Physics-Informed Neural Networks have shown unique utility in parameterising the solution of a well-defined partial differential equation using automatic differentiation and residual losses. Though they provide theoretical guarantees of convergence, in practice the required training regimes tend to be exacting and demanding. Through the course of this paper, we take a deep dive into understanding the loss landscapes associated with a PINN and how that offers some insight as to why PINNs are fundamentally hard to optimise for. We demonstrate how PINNs can be forced to converge better towards the solution, by way of feeding in sparse or coarse data...

[73] [Ensemble models outperform single model uncertainties and predictions for operator-learning of hypersonic flows](#)

Victor J. Leon, ..., and Alexander New. ArXiv, 2023. 0 citations.

2% Topic Match

Abstract: High-fidelity computational simulations and physical experiments of hypersonic flows are resource intensive. Training scientific machine learning (SciML) models on limited high-fidelity data offers one approach to rapidly predict behaviors for situations that have not been seen before. However, high-fidelity data is itself in limited quantity to validate all outputs of the SciML model in unexplored input space. As such, an uncertainty-aware SciML model is desired. The SciML model's output uncertainties could then be used to assess the reliability and confidence of the model's predictions. In this study, we extend a DeepONet using three different uncertainty quantification mechanisms: mean-variance estimation, evidential...

[74] [Benchmarking neural surrogates on realistic spatiotemporal multiphysics flows](#)

Runze Mao, ..., and Zhi X. Chen. ArXiv, 2025. 1 citations.

2% Topic Match

Abstract: Predicting multiphysics dynamics is computationally expensive and challenging due to the severe coupling of multi-scale, heterogeneous physical processes. While neural surrogates promise a paradigm shift, the field currently suffers from an "illusion of mastery", as repeatedly emphasized in top-tier commentaries: existing evaluations overly rely on simplified, low-dimensional proxies, which fail to expose the models' inherent fragility in realistic regimes. To bridge this critical gap, we present REALM (REalistic AI Learning for Multiphysics), a rigorous benchmarking framework designed to test neural surrogates on challenging, application-driven reactive flows. REALM features 11 high-fidelity datasets spanning from canonical multiphysics problems to complex propulsion and fire safety...

[75] [Valid \(Re-\)Use of Models-of-the-Physics in Cyber-Physical Systems Using Validity Frames](#)

Bert Van Acker, ..., and Kris Vanstechelma. 2019 Spring Simulation Conference (SpringSim), 2019. 16 citations.

2% Topic Match

Abstract: Modelling and Simulation techniques play a key role in the design of Software-Intensive and Cyber-Physical Systems. They enable not only early virtual integration of the different parts of the system but also the use of the dynamic behavior of the physical system in the control algorithms, to cope with the continuous demand for improved system control and performance. These physics-based high-fidelity models are mostly numerically-intensive that cannot be deployed on restricted embedded targets without altering the model. Altering these high-fidelity models, e.g. by performing mathematical reductions, decreases the model(re-)usability if the range-of-validity of this model is not formally captured. In...

[76] [Benchmarking Long Roll-outs of Auto-regressive Neural Operators for the Compressible Navier-Stokes Equations with Conserved Quantity Correction](#)

Sean Current, ..., and Srinivasan Parthasarathy. ArXiv, 2026. 0 citations.

1% Topic Match

Abstract: Deep learning has been proposed as an efficient alternative for the numerical approximation of PDE solutions, offering fast, iterative simulation of PDEs through the approximation of solution operators. However, deep learning solutions have struggle to perform well over long prediction durations due to the accumulation of auto-regressive error, which is compounded by the inability of models to conserve physical quantities. In this work, we present conserved quantity correction, a model-agnostic technique for incorporation physical conservation criteria within deep learning models. Our results demonstrate consistent improvement in the long-term stability of auto-regressive neural operator models, regardless of the model architecture. Furthermore,...

[77] [Data-Efficient Neural Operator Training via Physics-Based Active Learning](#)

Alicja Polanska, ..., and Stanislas Pamela. ArXiv, 2026. 0 citations.

1% Topic Match

Abstract: Solving partial differential equations with neural operators significantly reduces computational costs but remains bottlenecked by high training data requirements. Active learning offers a natural framework to mitigate this by selectively acquiring the most informative samples in an iterative manner. We introduce physics-based acquisition - a novel physics-informed active learning algorithm that leverages the partial differential equation residual to guide data selection. We validate the method by presenting numerical experiments for the 1D Burgers equation and the 2D compressible Navier-Stokes equations. We show that, in our experiments, physics-based acquisition consistently outperforms random acquisition and matches the state of the art in...

[78] [Signal-Aware Conditional Diffusion Surrogates for Transonic Wing Pressure Prediction](#)

Victor Franc'es-Belda, ..., and Rodrigo Castellanos. ArXiv, 2026. 0 citations.

1% Topic Match

Abstract: Accurate and efficient surrogate models for aerodynamic surface pressure fields are essential for accelerating aircraft design and analysis, yet deterministic regressors trained with pointwise losses often smooth sharp nonlinear features. This work presents a conditional denoising diffusion probabilistic model for predicting surface pressure distributions on the NASA Common Research Model wing under varying conditions of Mach number, angle of attack, and four control surface deflections. The framework operates on unstructured surface data through a principal component representation used as a non-truncated, reversible linear reparameterization of the pressure field, enabling a fully connected architecture. A signal-aware training objective is derived by...

[79] [Hidden physics models: Machine learning of nonlinear partial differential equations](#)

M. Raissi and G. Karniadakis. J. Comput. Phys., 2017. 1326 citations.

1% Topic Match

Abstract: Abstract While there is currently a lot of enthusiasm about "big data", useful data is usually "small" and expensive to acquire. In this paper, we present a new paradigm of learning partial differential equations from small data. In particular, we introduce hidden physics models, which are essentially data-efficient learning machines capable of leveraging the underlying laws of physics, expressed by time dependent and nonlinear partial differential equations, to extract patterns from high-dimensional data generated from experiments. The proposed methodology may be applied to the problem of learning, system identification, or data-driven discovery of partial differential equations. Our framework relies on...

[80] [Artificial Intelligence-Driven Surrogate Modelling for Nozzle Flow Simulation: A Review of Methods, Applications, and Challenges](#)

Usman Abdulrahman Sunday, ..., and Muhammad Abubakar Alhassan. European Journal of Computer Science and Information Technology, 2026. 0 citations.

1% Topic Match

Abstract: This review explores the transformative role of artificial intelligence in surrogate modelling for nozzle flow simulations, addressing the computational bottlenecks associated with high-fidelity computational fluid dynamics. Focusing on post-2020 advancements, the paper categorizes key methodologies into physics-based surrogates (such as Kriging and Radial Basis Functions), pure data-driven AI methods (including CNN-LSTM and Gaussian Processes), and AI-enhanced reduced-order modelling, such as Physics-Informed Neural Networks. Prominent applications are examined, ranging from high-precision flow rate prediction in sonic nozzles, achieving root-mean-square errors as low as 0.17%, to thrust optimization in aerospace components and real-time control of cold gas propulsion systems. While specific industrial...

[81] [Evaluating Uncertainty Quantification approaches for Neural PDEs in scientific applications](#)

Vardhan Dongre and G. S. Hora. ArXiv, 2023. 0 citations.

1% Topic Match

Abstract: The accessibility of spatially distributed data, enabled by affordable sensors, field, and numerical experiments, has facilitated the development of data-driven solutions for scientific problems, including climate change, weather prediction, and urban planning. Neural Partial Differential Equations (Neural PDEs), which combine deep learning (DL) techniques with domain expertise (e.g., governing equations) for parameterization, have proven to be effective in capturing valuable correlations within spatiotemporal datasets. However, sparse and noisy measurements coupled with modeling approximation introduce aleatoric and epistemic uncertainties. Therefore, quantifying uncertainties propagated from model inputs to outputs remains a challenge and an essential goal for establishing the trustworthiness of Neural...

[82] [Active Learning for Neural PDE Solvers](#)

Daniel Musekamp, ..., and Mathias Niepert. ArXiv, 2024. 26 citations.

1% Topic Match

Abstract: Solving partial differential equations (PDEs) is a fundamental problem in science and engineering. While neural PDE solvers can be more efficient than established numerical solvers, they often require large amounts of training data that is costly to obtain. Active learning (AL) could help surrogate models reach the same accuracy with smaller training sets by querying classical solvers with more informative initial conditions and PDE parameters. While AL is more common in other domains, it has yet to be studied extensively for neural PDE solvers. To bridge this gap, we introduce AL4PDE, a modular and extensible active learning benchmark. It provides...

[83] [Quantifying model form uncertainty in Reynolds-averaged turbulence models with Bayesian deep neural networks](#)

N. Geneva and N. Zabaras. J. Comput. Phys., 2018. 139 citations.

1% Topic Match

Abstract: Abstract Data-driven methods for improving turbulence modeling in Reynolds-Averaged Navier–Stokes (RANS) simulations have gained significant interest in the computational fluid dynamics community. Modern machine learning algorithms have opened up a new area of black-box turbulence models allowing for the tuning of RANS simulations to increase their predictive accuracy. While several data-driven turbulence models have been reported, the quantification of the uncertainties introduced has mostly been neglected. Uncertainty quantification for such data-driven models is essential since their predictive capability rapidly declines as they are tested for flow physics that deviate from that in the training data. In this work, we propose...

[84] [Diagnosing Failure Modes of Neural Operators Across Diverse PDE Families](#)

Lennon Shikhan. ArXiv, 2026. 4 citations.

1% Topic Match

Abstract: Neural PDE solvers are increasingly used as learned surrogates for families of partial differential equations, where the key machine learning challenge is not only interpolation on a fixed benchmark distribution but generalization under structured shifts in coefficients, boundary conditions, discretization, and rollout horizon. Yet evaluation is still often dominated by in-distribution test error, making robustness difficult to assess. We introduce a standardized stress-testing framework for neural PDE solvers under deployment-relevant shift. We instantiate it on three representative architectures -- Fourier Neural Operators (FNOs), a DeepONet-style model, and convolutional neural operators (CNOs) -- across five qualitatively different PDE families: dispersive, elliptic,...

[85] [Developing block-based physics-informed multi-layered neural network model for simulating the inelastic response of base-isolated structures](#)

A. Habib and U. Yildirim. Neural Computing and Applications, 2025. 5 citations.

1% Topic Match

No summary or abstract available

[86] [Scientific Machine Learning Through Physics-Informed Neural Networks: Where we are and What's Next](#)

S. Cuomo, ..., and F. Piccialli. Journal of Scientific Computing, 2022. 2394 citations.

1% Topic Match

Abstract: Physics-Informed Neural Networks (PINN) are neural networks (NNs) that encode model equations, like Partial Differential Equations (PDE), as a component of the neural network itself. PINNs are nowadays used to solve PDEs, fractional equations, integral-differential equations, and stochastic PDEs. This novel methodology has arisen as a multi-task learning framework in which a NN must fit observed data while reducing a PDE residual. This article provides a comprehensive review of the literature on PINNs: while the primary goal of the study was to characterize these networks and their related advantages and disadvantages. The review also attempts to incorporate publications on a...

[87] [StablePDENet: Enhancing Stability of Operator Learning for Solving Differential Equations](#)

Chutian Huang, ..., and Yang Xiang. ArXiv, 2026. 0 citations.

1% Topic Match

Abstract: Learning solution operators for differential equations with neural networks has shown great potential in scientific computing, but ensuring their stability under input perturbations remains a critical challenge. This paper presents a robust self-supervised neural operator framework that enhances stability through adversarial training while preserving accuracy. We formulate operator learning as a min-max optimization problem, where the model is trained against worst-case input perturbations to achieve consistent performance under both normal and adversarial conditions. We demonstrate that our method not only achieves good performance on standard inputs, but also maintains high fidelity under adversarial perturbed inputs. The results highlight the importance...

[88] [PINNfluence: Interpreting PINNs through Influence Functions](#)

Aleksander Krasowski, ..., and R. P. Klausen. ArXiv, 2024. 0 citations.

1% Topic Match

Abstract: Physics-informed neural networks (PINNs) have emerged as a powerful deep learning approach for solving partial differential equations (PDEs) in the physical sciences, yet their behavior remains largely opaque and is typically understood through failure mode analyses rather than explicit interpretability. To address this issue, we introduce PINNfluence, a training data attribution framework for interpreting PINNs based on influence functions. By extending influence functions to composite physics-informed training objectives, we enable fine-grained attribution between predictions, loss components, and training data points. Through benchmark experiments across various PDEs, we demonstrate that influence patterns provide granular diagnostics that distinguish structural characteristics across well-trained...

[89] [Representation Equivalent Neural Operators: a Framework for Alias-free Operator Learning](#)

Francesca Bartolucci, ..., and Rima Alaifari. Advances in Neural Information Processing Systems 36, 2023. 68 citations.

1% Topic Match

Abstract: Recently, operator learning, or learning mappings between infinite-dimensional function spaces, has garnered significant attention, notably in relation to learning partial differential equations from data. Conceptually clear when outlined on paper, neural operators necessitate discretization in the transition to computer implementations. This step can compromise their integrity, often causing them to deviate from the underlying operators. This research offers a fresh take on neural operators with a framework Representation equivalent Neural Operators (ReNO) designed to address these issues. At its core is the concept of operator aliasing, which measures inconsistency between neural operators and their discrete representations. We explore this for...

[90] [Surrogate Modelling and Uncertainty Quantification Based on Multi-Fidelity Deep Neural Network](#)

Zhihui Li and F. Montomoli. SSRN Electronic Journal, 2023. 21 citations.

1% Topic Match

Abstract: To reduce training costs, several Deep neural networks (DNNs) that can learn from a small set of HF data and a sufficient number of low-fidelity (LF) data have been proposed. In these established neural networks, a parallel structure is commonly proposed to separately approximate the non-linear and linear correlation between the HF- and LF data. In this paper, a new architecture of multi-fidelity deep neural network (MF-DNN) was proposed where one subnetwork was built to approximate both the non-linear and linear correlation simultaneously. Rather than manually allocating the output weights for the paralleled linear and nonlinear correction networks, the proposed...

[91] [Generalization Bounds for Physics-Informed Neural Networks for the Incompressible Navier-Stokes Equations](#)

Sebastien Andre-Sloan, ..., and Anirbit Mukherjee. ArXiv, 2026. 0 citations.

1% Topic Match

Abstract: This work establishes rigorous first-of-its-kind upper bounds on the generalization error for the method of approximating solutions to the (d+1)-dimensional incompressible Navier-Stokes equations by training depth-2 neural networks trained via the unsupervised Physics-Informed Neural Network (PINN) framework. This is achieved by bounding the Rademacher complexity of the PINN risk. For appropriately weight bounded net classes our derived generalization bounds do not explicitly depend on the network width and our framework characterizes the generalization gap in terms of the fluid's kinematic viscosity and loss regularization parameters. In particular, the resulting sample complexity bounds are dimension-independent. Our generalization bounds suggest using novel...

[92] [Uncertainty-Aware Diagnostics for Physics-Informed Machine Learning](#)

M. Daniels, ..., and Michael W. Mahoney. ArXiv, 2025. 2 citations.

1% Topic Match

Abstract: Physics-informed machine learning (PIML) integrates prior physical information, often in the form of differential equation constraints, into the process of fitting machine learning models to physical data. Popular PIML approaches, including neural operators, physics-informed neural networks, neural ordinary differential

equations, and neural discrete equilibria, are typically fit to objectives that simultaneously include both data and physical constraints. However, the multi-objective nature of this approach creates ambiguity in the measurement of model quality. This is related to a poor understanding of epistemic uncertainty, and it can lead to surprising failure modes, even when existing statistical metrics suggest strong fits. Working within...

[93] [Fourier Neural Operator for Plasma Modelling](#)

Vignesh Gopakumar, ..., and Mast Team. ArXiv, 2023. 31 citations.

1% Topic Match

Abstract: Predicting plasma evolution within a Tokamak is crucial to building a sustainable fusion reactor. Whether in the simulation space or within the experimental domain, the capability to forecast the spatio-temporal evolution of plasma field variables rapidly and accurately could improve active control methods on current tokamak devices and future fusion reactors. In this work, we demonstrate the utility of using Fourier Neural Operator (FNO) to model the plasma evolution in simulations and experiments. Our work shows that the FNO is capable of predicting magnetohydrodynamic models governing the plasma dynamics, 6 orders of magnitude faster than the traditional numerical solver, while...

[94] [Artificial neural networks for solving ordinary and partial differential equations](#)

I. Lagaris, ..., and D. Fotiadis. IEEE transactions on neural networks, 1997. 2694 citations.

1% Topic Match

Abstract: We present a method to solve initial and boundary value problems using artificial neural networks. A trial solution of the differential equation is written as a sum of two parts. The first part satisfies the initial/boundary conditions and contains no adjustable parameters. The second part is constructed so as not to affect the initial/boundary conditions. This part involves a feedforward neural network containing adjustable parameters (the weights). Hence by construction the initial/boundary conditions are satisfied and the network is trained to satisfy the differential equation. The applicability of this approach ranges from single ordinary differential equations (ODE's), to systems of...

[95] [PhysicsCorrect: A Training-Free Approach for Stable Neural PDE Simulations](#)

Xinquan Huang and P. Perdikaris. Unknown journal, 2025. 5 citations.

1% Topic Match

Abstract: Neural networks have emerged as powerful surrogates for solving partial differential equations (PDEs), offering significant computational speedups over traditional methods. However, these models suffer from a critical limitation: error accumulation during long-term rollouts, where small inaccuracies compound exponentially, eventually causing complete divergence from physically valid solutions. We present PhysicsCorrect, a training-free correction framework that enforces PDE consistency at each prediction step by formulating correction as a linearized inverse problem based on PDE residuals. Our key innovation is an efficient caching strategy that precomputes the Jacobian and its pseudoinverse during an offline warm-up phase, reducing computational overhead by two orders of...

[96] [Application of deep learning based multi-fidelity surrogate model to robust aerodynamic design optimization](#)

Jun Tao and Gang Sun. Aerospace Science and Technology, 2019. 193 citations.

1% Topic Match

Abstract: Abstract In the present work, a multi-fidelity surrogate-based optimization framework is proposed, and then applied to the robust optimizations for airfoil and wing under uncertainty of Mach number. DBN (deep belief network) is employed as the low-fidelity model, and the k-step contrastive divergence algorithm is used for training the network. By virtue of the well trained DBN model and high-fidelity data, a linear regression multi-fidelity surrogate model is established. Verification results indicate that the multi-fidelity surrogate model obtains more accurate predictions than the DBN model and is highly reliable as a prediction model. The multi-fidelity surrogate model is embedded...

[97] [Linking Physical Fidelity to Downstream Performance in Physics-Informed Fault Diagnosis](#)

E. Spotorno, ..., and Antônio Augusto Fröhlich. IEEE Access, 2026. 0 citations.

1% Topic Match

Abstract: A critical question in Scientific Machine Learning is whether improving a Physics-Informed Neural Network's (PINN) physical fidelity translates into downstream task performance. We investigate this link using a three-stage diagnostic framework: PINN-based residual generation, Extreme Value Theory anomaly detection, and Siamese Neural Network (SNN) open-set recognition. To overcome optimization costs, we propose a hierarchical three-phase evaluation protocol that progressively filters models based on physical fidelity (Phase 1), linear separability (Phase 2), and downstream robustness (Phase 3). Applying this to a ring bearing system, we evaluated adaptive PINNs against rigorous, architecturally identical data-driven baselines. Results revealed a paradox: unconstrained baselines achieved...

[98] [Fourier Neural Operator for Parametric Partial Differential Equations](#)

Zong-Yi Li, ..., and Anima Anandkumar. ArXiv, 2020. 4314 citations.

0% Topic Match

Abstract: The classical development of neural networks has primarily focused on learning mappings between finite-dimensional Euclidean spaces. Recently, this has been generalized to neural operators that learn mappings between function spaces. For partial differential equations (PDEs), neural operators directly learn the mapping from any functional parametric dependence to the solution. Thus, they learn an entire family of PDEs, in contrast to classical methods which solve one instance of the equation. In this work, we formulate a new neural operator by parameterizing the integral kernel directly in Fourier space, allowing for an expressive and efficient architecture. We perform experiments on Burgers' equation,...

[99] [Plasma surrogate modelling using Fourier neural operators](#)

Vignesh Gopakumar, ..., and Mast Team. Nuclear Fusion, 2023. 53 citations.

0% Topic Match

Abstract: Predicting plasma evolution within a Tokamak reactor is crucial to realizing the goal of sustainable fusion. Capabilities in forecasting the spatio-temporal evolution of plasma rapidly and accurately allow us to quickly iterate over design and control strategies on current Tokamak devices and future reactors. Modelling plasma evolution using numerical solvers is often expensive, consuming many hours on supercomputers, and hence, we need alternative inexpensive surrogate models. We demonstrate accurate predictions of plasma evolution both in simulation and experimental domains using deep learning-based surrogate modelling tools, viz., Fourier neural operators (FNO). We show that FNO has a speedup of six orders...

[100] [Neural-network methods for boundary value problems with irregular boundaries](#)

I. Lagaris, ..., and D. Papageorgiou. IEEE transactions on neural networks, 2000. 537 citations.

0% Topic Match

No summary or abstract available

[101] [Neural network surrogates with uncertainty quantification for inverse problems in partial differential equations](#)

Christian Jimenez-Beltran, ..., and Konstantinos C. Zygalakis. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Inverse problems for differential equations arise throughout science and engineering, where one seeks to infer unknown model parameters from noisy or incomplete observations. Traditional numerical methods for these problems are often computationally expensive, particularly in Bayesian settings where evaluating the likelihood becomes costly for complex forward models and high-dimensional parameter spaces. To address this challenge, we introduce DeepGaLA, a neural-network surrogate for differential equation solvers that provides uncertainty-aware predictions, reducing overconfident inference when training data are limited. To evaluate the fidelity of the surrogate-induced posterior approximations in practice, we show that a short run of delayed-acceptance Markov chain Monte Carlo...

[102] [Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators](#)

Lu Lu, ..., and G. Karniadakis. Nature Machine Intelligence, 2019. 3910 citations.

0% Topic Match

Abstract: It is widely known that neural networks (NNs) are universal approximators of continuous functions. However, a less known but powerful result is that a NN with a single hidden layer can accurately approximate any nonlinear continuous operator. This universal approximation theorem of operators is suggestive of the structure and potential of deep neural networks (DNNs) in learning continuous operators or complex systems from streams of scattered data. Here, we thus extend this theorem to DNNs. We design a new network with small generalization error, the deep operator network (DeepONet), which consists of a DNN for encoding the discrete input function...

[103] [Invariant preservation in machine learned PDE solvers via error correction](#)

Nick McGreivy and Ammar Hakim. ArXiv, 2023. 11 citations.

0% Topic Match

Abstract: Machine learned partial differential equation (PDE) solvers trade the reliability of standard numerical methods for potential gains in accuracy and/or speed. The only way for a solver to guarantee that it outputs the exact solution is to use a convergent method in the limit that the grid spacing  $\Delta x$  and timestep  $\Delta t$  approach zero. Machine learned solvers, which learn to update the solution at large  $\Delta x$  and/or  $\Delta t$ , can never guarantee perfect accuracy. Some amount of error is inevitable, so the question becomes: how do we constrain machine learned solvers to give us the sorts of...

[104] [Neural networks for rarefied gas dynamics: Relaxation problem, polyatomic shock waves, and hypersonic cylinder flow](#)

E. Roohi, ..., and F. E. Azghadi. Physics of Fluids, 2026. 4 citations.

0% Topic Match

Abstract: This work presents a suite of targeted methodologies based on neural operators, including physics-informed, physics-constrained, and data-driven approaches, for constructing computationally efficient and robust surrogate models in rarefied gas dynamics, where high-fidelity kinetic solvers are often prohibitively expensive. Three key contributions are introduced, demonstrating stability, physics-discovery capability, and strong generalization across different regimes. First, for the Bhatnagar–Gross–Krook kinetic relaxation problem, a perturbation ansatz is proposed to ensure numerical stability within a physics-informed neural network framework. This stabilized model enables simultaneous forward prediction and decay-rate identification, successfully inferring the unknown collision frequency solely from the governing equations and initial conditions. Validation...

[105] [Neural-network-based approximations for solving partial differential equations](#)

M. Dissanayake and N. Phan-Thien. Communications in Numerical Methods in Engineering, 1994. 645 citations.

0% Topic Match

No summary or abstract available

[106] [Physics-Constrained Deep Learning for High-dimensional Surrogate Modeling and Uncertainty Quantification without Labeled Data](#)

Yinhao Zhu, ..., and P. Perdikaris. J. Comput. Phys., 2019. 1037 citations.

0% Topic Match

Abstract: Abstract Surrogate modeling and uncertainty quantification tasks for PDE systems are most often considered as supervised learning problems where input and output data pairs are used for training. The construction of such emulators is by definition a small data problem which poses challenges to deep learning approaches that have been developed to operate in the big data regime. Even in cases where such models have been shown to have good predictive capability in high dimensions, they fail to address constraints in the data implied by the PDE model. This paper provides a methodology that incorporates the governing equations of the...

[107] [Hybrid Neural World Models](#)

Pranav Lakshmanan and Paras Chopra. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Neural surrogates promise large speedups over classical solvers for physical dynamics but fail silently at sharp dynamical events such as shocks, fronts, and contact. We present hybrid neural world models for physical dynamics: a recipe for training and deploying multi-horizon surrogates in physical state space, where a single network with continuous horizon conditioning is trained with direct supervision against textbook reference solvers to predict any future state at horizon  $T$  in one forward pass. Although no part of the training data, loss function, or architecture supervises discontinuity location, the trained surrogate encodes it implicitly, recoverable from its forward passes alone...

[108] [PDE-SHARP: PDE Solver Hybrids through Analysis and Refinement Passes](#)

Shaghayegh Fazliani and Madeleine Udell. ArXiv, 2025. 2 citations.

0% Topic Match

Abstract: Current LLM-driven approaches using test-time computing to generate PDE solvers execute a large number of solver samples to identify high-accuracy solvers. These paradigms are especially costly for complex PDEs requiring substantial computational resources for numerical evaluation. We introduce PDE-SHARP, a framework to reduce computational costs by replacing expensive scientific computation by cheaper LLM inference that achieves superior solver accuracy with 60-75% fewer computational evaluations. PDE-SHARP employs three stages: (1) Analysis: mathematical chain-of-thought analysis including PDE classification, solution type detection, and stability analysis; (2) Genesis: solver generation based on mathematical insights from the previous stage; and (3) Synthesis: collaborative selection-hybridization tournaments...

[109] [Physics Enhanced Deep Surrogates for the Phonon Boltzmann Transport Equation](#)

Antonio Varagnolo, ..., and Raphael Pestourie. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: Designing materials with controlled heat flow at the nano-scale is central to advances in microelectronics, thermoelectrics, and energy-conversion technologies. At these scales, phonon transport follows the Boltzmann Transport Equation (BTE), which captures non-diffusive (ballistic) effects but is too costly to solve repeatedly in inverse-design loops. Existing surrogate approaches trade speed for accuracy: fast macroscopic solvers can overestimate conductivities by hundreds of percent, while recent data-driven operator learners often require thousands of high-fidelity simulations. This creates a need for a fast, data-efficient surrogate that remains reliable across ballistic and diffusive regimes. We introduce a Physics-Enhanced Deep Surrogate (PEDS) that combines...

[110] [Biophysics Informed Pathological Regularisation for Brain Tumour Segmentation](#)

Lipei Zhang, ..., and Angelica I. Avilés-Rivero. Unknown journal, 2024. 13 citations.

0% Topic Match

Abstract: Recent advances in deep learning have significantly improved brain tumour segmentation techniques; however, the results still lack confidence and robustness as they solely consider image data without biophysical priors or pathological information. Integrating biophysics-informed regularisation is one effective way to change this situation, as it provides an prior regularisation for automated end-to-end learning. In this paper, we propose a novel approach that designs brain tumour growth Partial Differential Equation (PDE) models as a regularisation with deep learning, operational with any network model. Our method introduces tumour growth PDE models directly into the segmentation process, improving accuracy and robustness, especially in...

[111] [Unified theoretical guarantees for stability, consistency, and convergence in neural PDE solvers from non-IID data to physics-informed networks](#)

Ronald Katende. ArXiv, 2024. 0 citations.

0% Topic Match

Abstract: We establish a unified theoretical framework addressing the stability, consistency, and convergence of neural networks under realistic training conditions, specifically, in the presence of non-IID data, geometric constraints, and embedded physical laws. For standard supervised learning with dependent data, we derive uniform stability bounds for gradient-based methods using mixing coefficients and dynamic learning rates. In federated learning with heterogeneous data and non-Euclidean parameter spaces, we quantify model inconsistency via curvature-aware aggregation and information-theoretic divergence. For Physics-Informed Neural Networks (PINNs), we rigorously prove perturbation stability, residual consistency, Sobolev convergence, energy stability for conservation laws, and convergence under adaptive multi-domain refinements. Each...

[112] [Learning Under Laws: A Constraint-Projected Neural PDE Solver that Eliminates Hallucinations](#)

Mainak Singha. ArXiv, 2025. 1 citations.

0% Topic Match

Abstract: Neural networks can approximate solutions to partial differential equations, but they often break the very laws they are meant to model—creating mass from nowhere, drifting shocks, or violating conservation and entropy. We address this by training within the laws of physics rather than beside them. Our framework, called Constraint-Projected Learning (CPL), keeps every update physically admissible by projecting network outputs onto the intersection of constraint sets defined by conservation, Rankine-Hugoniot balance, entropy, and positivity. The projection is differentiable and adds only about 10% computational overhead, making it fully compatible with back-propagation. We further stabilize training with total-variation damping (TVD) to...

[113] [Scaling physics-informed hard constraints with mixture-of-experts](#)

N. Chalapathi, ..., and Aditi Krishnapriyan. ArXiv, 2024. 32 citations.

0% Topic Match

Abstract: Imposing known physical constraints, such as conservation laws, during neural network training introduces an inductive bias that can improve accuracy, reliability, convergence, and data efficiency for modeling physical dynamics. While such constraints can be softly imposed via loss function penalties, recent advancements in differentiable physics and optimization improve performance by incorporating PDE-constrained optimization as individual layers in neural networks. This enables a stricter adherence to physical constraints. However, imposing hard constraints significantly increases computational and memory costs, especially for complex dynamical systems. This is because it requires solving an optimization problem over a large number of points in a mesh,...

[114] [Bias Inheritance in Neural-Symbolic Discovery of Constitutive Closures Under Function-Class Mismatch](#)

Han-Yu Liang, ..., and Fujun Liu. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: We investigate the data-driven discovery of constitutive closures in nonlinear reaction-diffusion systems with known governing PDE structures. Our objective is to robustly recover diffusion and reaction laws from spatiotemporal observations while avoiding the common pitfall where low residuals or short-horizon predictions are conflated with physical recovery. We propose a three-stage neural-symbolic framework: (1) learning numerical surrogates under physical constraints using a noise-robust weak-form-driven objective; (2) compressing these surrogates into restricted interpretable symbolic families (e.g., polynomial, rational, and saturation forms); and (3) validating the symbolic closures through explicit forward re-simulation on unseen initial conditions. Extensive numerical experiments reveal two distinct regimes....

[115] [Calibrated Uncertainty Quantification for Operator Learning via Conformal Prediction](#)

Ziqi Ma, ..., and Anima Anandkumar. Trans. Mach. Learn. Res., 2024. 20 citations.

0% Topic Match

Abstract: Operator learning has been increasingly adopted in scientific and engineering applications, many of which require calibrated uncertainty quantification. Since the output of operator learning is a continuous function, quantifying uncertainty simultaneously at all points in the domain is challenging. Current methods consider calibration at a single point or over one scalar function or make strong assumptions such as Gaussianity. We propose a risk-controlling quantile neural operator, a distribution-free, finite-sample functional calibration conformal prediction method. We provide a theoretical calibration guarantee on the coverage rate, defined as the expected percentage of points on the function domain whose true value lies within...

[116] [Convergence Guarantees for Gradient-Based Training of Neural PDE Solvers: From Linear to Nonlinear PDEs](#)

Wei Zhao and Tao Luo. ArXiv, 2025. 1 citations.

0% Topic Match

Abstract: We present a unified convergence theory for gradient-based training of neural network methods for partial differential equations (PDEs), covering both physics-informed neural networks (PINNs) and the Deep Ritz method. For linear PDEs, we extend the neural tangent kernel (NTK) framework for PINNs to establish global convergence guarantees for a broad class of linear operators. For nonlinear PDEs, we prove convergence to critical points via the Łojasiewicz inequality under the random feature model, eliminating the need for strong over-parameterization and encompassing both gradient flow and implicit gradient descent dynamics. Our results further reveal that the random feature model exhibits an implicit...

[117] [A Conformal Prediction Framework for Uncertainty Quantification in Physics-Informed Neural Networks](#)

Yifan Yu, ..., and Yangshuai Wang. J. Comput. Phys., 2025. 4 citations.

0% Topic Match

Abstract: Physics-Informed Neural Networks (PINNs) have emerged as a powerful framework for solving PDEs, yet existing uncertainty quantification (UQ) approaches for PINNs generally lack rigorous statistical guarantees. In this work, we bridge this gap by introducing a distribution-free conformal prediction (CP) framework for UQ in PINNs. This framework calibrates prediction intervals by constructing nonconformity scores on a calibration set, thereby yielding distribution-free uncertainty estimates with rigorous finite-sample coverage guarantees for PINNs. To handle spatial heteroskedasticity, we further introduce local conformal quantile estimation, enabling spatially adaptive uncertainty bands while preserving theoretical guarantee. Through systematic evaluations on typical PDEs (damped harmonic oscillator, Poisson,...

[118] [Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations](#)

M. Raissi, ..., and G. Karniadakis. J. Comput. Phys., 2019. 17735 citations.

0% Topic Match

Abstract: Abstract We introduce physics-informed neural networks – neural networks that are trained to solve supervised learning tasks while respecting any given laws of physics described by general nonlinear partial differential equations. In this work, we present our developments in the context of solving two main classes of problems: data-driven solution and data-driven discovery of partial differential equations. Depending on the nature and arrangement of the available data, we devise two distinct types of algorithms, namely continuous time and discrete time models. The first type of models forms a new family of data-efficient spatio-temporal function approximators, while the latter type allows...

[119] [Evaluation of machine learning 1 architectures on the quantification of 2 epistemic and aleatoric uncertainties in 3 complex dynamical systems](#)

S. Guth, ..., and T. Sapsis. Journal Not Provided, 2023. 2 citations.

0% Topic Match

No summary or abstract available

[120] [Conformal Prediction for Neural Operators: Distribution-Free Uncertainty Quantification in Physics Simulation](#)

Michael Chin. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Neural operators such as the Fourier Neural Operator (FNO) have emerged as powerful surrogates for solving partial differential equations (PDEs), achieving speedups of several orders of magnitude over traditional numerical solvers. However, deploying these models in safety-critical engineering applications -- such as thermal management of electronic components and battery systems -- requires not only accurate point predictions but also rigorous uncertainty guarantees. Existing uncertainty quantification (UQ) methods for neural operators, including Monte Carlo Dropout and Deep Ensembles, provide only relative uncertainty estimates without formal coverage guarantees. In this work, we propose the first application of split conformal prediction to neural...

[121] [Distribution-Free Robust Predict-Then-Optimize in Function Spaces](#)

Yash Patel and Ambuj Tewari. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: The need to rapidly solve PDEs in engineering design workflows has spurred the rise of neural surrogate models. In particular, neural operator models provide a discretization-invariant surrogate by retaining the infinite-dimensional, functional form of their arguments. Despite improved throughput, such methods lack guarantees on accuracy, unlike classical numerical PDE solvers. Optimizing engineering designs under these potentially miscalibrated surrogates thus runs the risk of producing designs that perform poorly upon deployment. In a similar vein, there is growing interest in automated decision-making under black-box predictors in the finite-dimensional setting, where a similar risk of suboptimality exists under poorly calibrated models. For...

[122] [Physics-informed machine learning](#)

G. Karniadakis, ..., and Liu Yang. Nature Reviews Physics, 2021. 7061 citations.

0% Topic Match

No summary or abstract available

[123] [Variational Physics-Informed Neural Networks For Solving Partial Differential Equations](#)

E. Kharazmi, ..., and G. Karniadakis. ArXiv, 2019. 333 citations.

0% Topic Match

Abstract: Physics-informed neural networks (PINNs) [31] use automatic differentiation to solve partial differential equations (PDEs) by penalizing the PDE in the loss function at a random set of points in the domain of interest. Here, we develop a Petrov-Galerkin version of PINNs based on the nonlinear approximation of deep neural networks (DNNs) by selecting the {lem trial space} to be the space of neural networks and the {lem test space} to be the space of Legendre polynomials. We formulate the {textit{variational residual}} of the PDE using the DNN approximation by incorporating the variational form of the problem into the loss function...

[124] [Machine learning solutions looking for PDE problems](#)

No author found. Nature Machine Intelligence, 2025. 7 citations.

0% Topic Match

No summary or abstract available

[125] [Valid Error Bars for Neural Weather Models using Conformal Prediction](#)

Vignesh Gopakumar, ..., and M. Deisenroth. ArXiv, 2024. 2 citations.

0% Topic Match

Abstract: Neural weather models have shown immense potential as inexpensive and accurate alternatives to physics-based models. However, most models trained to perform weather forecasting do not quantify the uncertainty associated with their forecasts. This limits the trust in the model and the usefulness of the forecasts. In this work we construct and formalise a conformal prediction framework as a post-processing method for estimating this uncertainty. The method is model-agnostic and gives calibrated error bounds for all variables, lead times and spatial locations. No modifications are required to the model and the computational cost is negligible compared to model training. We demonstrate...

[126] [Reachability Analysis for One Dimensional Linear Parabolic Equations](#)

Hoang-Dung Tran, ..., and Taylor T. Johnson. Unknown journal, 2018. 5 citations.

0% Topic Match

Abstract: Abstract Partial differential equations (PDEs) mathematically describe a wide range of phenomena such as fluid dynamics, or quantum mechanics. Although great achievements have been accomplished in the field of numerical methods for solving PDEs, from a safety verification (or falsification) perspective, methods are still needed to verify (or falsify) a system whose dynamics is specified as a PDE that may depend not only on space, but also on time. As many cyber-physical systems (CPS) involve sensing and control of physical phenomena modeled as PDEs, reachability analysis of PDEs provides novel methods for safety verification and falsification. As a first step...

[127] [Conformalized Quantum DeepONet Ensembles for Scalable Operator Learning with Distribution-Free Uncertainty](#)

Purav Matlia, ..., and Guang Lin. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Operator learning enables fast surrogate modeling of high-dimensional dynamical systems, but existing approaches face two fundamental limitations: quadratic inference complexity and unreliable uncertainty quantification in safety-critical settings. We propose Conformalized Quantum DeepONet Ensembles, a framework that addresses both challenges simultaneously. By leveraging Quantum Orthogonal Neural Networks (QOrthoNNs), we reduce operator inference complexity from  $O(n^2)$  to  $O(n)$ , enabling scalable evaluation over fine discretizations. To provide rigorous uncertainty quantification, we combine ensemble-based epistemic modeling with adaptive conformal prediction, yielding distribution-free coverage guarantees. A key challenge in ensembling is that naive parallelism scales hardware resources linearly with the number of models. We resolve...

[128] [A tutorial on conformal prediction](#)

G. Shafer and Vladimir Vovk. ArXiv, 2007. 1653 citations.

0% Topic Match

Abstract: Conformal prediction uses past experience to determine precise levels of confidence in new predictions. Given an error probability  $\epsilon$ , together with a method that makes a prediction  $w$  of a label  $y$ , it produces a set of labels, typically containing  $w$ , that also contains  $y$  with probability  $1 - \epsilon$ . Conformal prediction can be applied to any method for producing  $w$ : a nearest-neighbor method, a support-vector machine, ridge regression, etc. Conformal prediction is designed for an on-line setting in which labels are predicted successively, each one being revealed before the next is predicted. The most novel and valuable feature of...

[129] [Sobolev Neural Network With Residual Weighting as a Surrogate in Linear and Non-Linear Mechanics](#)

A. O. Kilicsoy, ..., and M. Faes. IEEE Access, 2024. 2 citations.

0% Topic Match

Abstract: Areas of computational mechanics such as uncertainty quantification and optimization usually involve repeated evaluation of numerical models that represent the behavior of engineering systems. In the case of complex non-linear systems however, these models tend to be expensive to evaluate, making surrogate models quite valuable. Artificial neural networks approximate systems very well by taking advantage of the inherent information of its given training data. In this context, this paper investigates the improvement of the training process by including sensitivity information, which are partial derivatives w.r.t. inputs, as outlined by Sobolev training. In computational mechanics, sensitivities can be applied to neural...

[130] [On universal approximation and error bounds for Fourier Neural Operators](#)

Nikola B. Kovachki, ..., and Siddhartha Mishra. ArXiv, 2021. 378 citations.

0% Topic Match

Abstract: Fourier neural operators (FNOs) have recently been proposed as an effective framework for learning operators that map between infinite-dimensional spaces. We prove that FNOs are universal, in the sense that they can approximate any continuous operator to desired accuracy. Moreover, we suggest a mechanism by which FNOs can approximate operators associated with PDEs efficiently. Explicit error bounds are derived to show that the size of the FNO, approximating operators associated with a Darcy type elliptic PDE and with the incompressible Navier-Stokes equations of fluid dynamics, only increases sub (log)-linearly in terms of the reciprocal of the error. Thus, FNOs are...

[131] [An RBC-MSUQ Framework for Red Blood Cell Morpho-Mechanics](#)

Shuo Wang, ..., and Tao Zhou. ArXiv, 2025. 0 citations.

0% Topic Match

Abstract: Characterizing the morpho-mechanical properties of red blood cells (RBCs) is crucial for understanding microvascular transport mechanisms and cellular pathophysiological processes, yet current computational models are constrained by multi-source uncertainties including cross-platform experimental discrepancies and parameter identification stochasticity. We present RBC-MSUQ, a novel multi-stage uncertainty quantification framework tailored for RBCs. It integrates hierarchical Bayesian inference with diverse experimental datasets, establishing prior distributions for RBC parameters via microscopic simulations and literature-derived data. A dynamic annealing technique defines stress-free baselines, while deep neural network surrogates, optimized through sensitivity analysis, achieve sub- $10^{-2}$  prediction errors for efficient simulation approximation. Its two-stage hierarchical inference architecture constrains...

[132] [Learning to Test: Physics-Informed Representation for Dynamical Instability Detection](#)

Minxing Zheng, ..., and Shixiang Zhu. ArXiv, 2026. 0 citations.

0% Topic Match

Abstract: Many safety-critical scientific and engineering systems evolve according to differential-algebraic equations (DAEs), where dynamical behavior is constrained by physical laws and admissibility conditions. In practice, these systems operate under stochastically varying environmental inputs, so stability is not a static property but must be reassessed as the context distribution shifts. Repeated large-scale DAE simulation, however, is computationally prohibitive in high-dimensional or real-time settings. This paper proposes a test-oriented learning framework for stability assessment under distribution shift. Rather than re-estimating physical parameters or repeatedly solving the underlying DAE, we learn a physics-informed latent representation of contextual variables that captures stability-relevant structure and...

[133] [Science and Statistics](#)

G. Box. Journal of the American Statistical Association, 1976. 2206 citations.

0% Topic Match

No summary or abstract available

[134] [Leveraging viscous Hamilton-Jacobi PDEs for uncertainty quantification in scientific machine learning](#)

Zongren Zou, ..., and G. Karniadakis. ArXiv, 2024. 13 citations.

0% Topic Match

Abstract: Uncertainty quantification (UQ) in scientific machine learning (SciML) combines the powerful predictive power of SciML with methods for quantifying the reliability of the learned models. However, two major challenges remain: limited interpretability and expensive training procedures. We provide a new interpretation for UQ problems by establishing a new theoretical connection between some Bayesian inference problems arising in SciML and viscous Hamilton-Jacobi partial differential equations (HJ PDEs). Namely, we show that the posterior mean and covariance can be recovered from the spatial gradient and Hessian of the solution to a viscous HJ PDE. As a first exploration of this connection, we...

[135] [Using neural networks for dimensioning and certification of mechanical systems: model precision and accuracy](#)

Y. El Assami and B. Gély. Proceedings of the Fourteenth International Conference on Computational Structures Technology, 2022. 0 citations.

0% Topic Match

Abstract: Neural networks show impressive performance in lots of domains to handle problems of high complexity. They are universal approximators and can, in principle, be used to learn any type of model. Their use would be of a great benefit as they can be intended to automate major tasks within an engineering project (such as system dimensioning, certification, and criteria verification). However, it is not yet customary to use these technics for lack of competitiveness against forward engineering calculations. One major issue is the robustness and the difficulty to ensure high precisions for deterministic predictions. In this work, we investigate the...

[136] [GMRES: a generalized minimal residual algorithm for solving nonsymmetric linear systems](#)

Y. Saad and M. Schultz. Siam Journal on Scientific and Statistical Computing, 1986. 12105 citations.

0% Topic Match

No summary or abstract available

[137] [Physics-Informed Latent Neural Operator for Real-time Predictions of Complex Physical Systems](#)

Sharmila Karumuri, ..., and Somdatta Goswami. ArXiv, 2025. 8 citations.

0% Topic Match

No summary or abstract available

[138] [A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification](#)

Anastasios Nikolas Angelopoulos and Stephen Bates. ArXiv, 2021. 1116 citations.

0% Topic Match

Abstract: Black-box machine learning models are now routinely used in high-risk settings, like medical diagnostics, which demand uncertainty quantification to avoid consequential model failures. Conformal prediction is a user-friendly paradigm for creating statistically rigorous uncertainty sets/intervals for the predictions of such models. Critically, the sets are valid in a distribution-free sense: they possess explicit, non-asymptotic guarantees even without distributional assumptions or model assumptions. One can use conformal prediction with any pre-trained model, such as a neural network, to produce sets that are guaranteed to contain the ground truth with a user-specified probability, such as 90%. It is easy-to-understand, easy-to-use, and general,...

[139] [Estimating the carbon footprint of computational fluid dynamics](#)

J. Horwitz. Physics of Fluids, 2024. 9 citations.

0% Topic Match

Abstract: Computational resources have grown exponentially in the past few decades. These machines make possible research and design in fields as diverse as medicine, astronomy, and engineering. Despite ever-increasing computational capabilities, direct simulation of complex systems has remained challenging owing to the degrees of freedom involved. At the cusp of exascale computing, high-resolution simulation of practical problems with minimal model assumptions may soon experience a renaissance. However, growing reliance on modern computers comes at the cost of a growing carbon footprint. To illustrate this, we examine historic computations in fluid dynamics where larger computers have afforded the opportunity to simulate flows...

[140] [A Review of Uncertainty Quantification in Deep Learning: Techniques, Applications and Challenges](#)

M. Abdar, ..., and S. Nahavandi. Inf. Fusion, 2020. 2624 citations.

0% Topic Match

Abstract: Uncertainty quantification (UQ) plays a pivotal role in reduction of uncertainties during both optimization and decision making processes. It can be applied to solve a variety of real-world applications in science and engineering. Bayesian approximation and ensemble learning techniques are two most widely-used UQ methods in the literature. In this regard, researchers have proposed different UQ methods and examined their performance in a variety of applications such as computer vision (e.g., self-driving cars and object detection), image processing (e.g., image restoration), medical image analysis (e.g., medical image classification and segmentation), natural language processing (e.g., text classification, social media texts and...

[141] [DGM: A deep learning algorithm for solving partial differential equations](#)

Justin A. Sirignano and K. Spiliopoulos. J. Comput. Phys., 2017. 2504 citations.

Not measured Topic Match

Abstract: High-dimensional PDEs have been a longstanding computational challenge. We propose a deep learning algorithm similar in spirit to Galerkin methods, using a deep neural network instead of linear combinations of basis functions. The PDE is approximated with a deep neural network, which is trained on random batches of spatial points to satisfy the differential operator and boundary conditions. The algorithm is mesh-less, which is key since meshes become infeasible in higher dimensions. Instead of forming a mesh, sequences of spatial points are randomly sampled. We implement the approach for American options (a type of free-boundary PDE which is widely used)...

[142] [Quasi-Optimal Least Squares: Inhomogeneous boundary conditions, and application with machine learning](#)

Harald Monsuur, ..., and Rob Stevenson. ArXiv, 2024. 2 citations.

Not measured Topic Match

Abstract: We construct least squares formulations of PDEs with inhomogeneous essential boundary conditions, where boundary residuals are not measured in unpractical fractional Sobolev norms, but which formulations nevertheless are shown to yield a quasi-best approximations from the employed trial spaces. Dual norms do enter the least-squares functional, so that solving the least squares problem amounts to solving a saddle point or minimax problem. For finite element applications we construct uniformly stable finite element pairs, whereas for Machine Learning applications we employ adversarial networks.

[143] [NSFnets \(Navier-Stokes flow nets\): Physics-informed neural networks for the incompressible Navier-Stokes equations](#)

Xiaowei Jin, ..., and G. Karniadakis. J. Comput. Phys., 2020. 1225 citations.

Not measured Topic Match

Abstract: We employ physics-informed neural networks (PINNs) to simulate the incompressible flows ranging from laminar to turbulent flows. We perform PINN simulations by considering two different formulations of the Navier-Stokes equations: the velocity-pressure (VP) formulation and the vorticity-velocity (VV) formulation. We refer to these specific PINNs for the Navier-Stokes flow nets as NSFnets. Analytical solutions and direct numerical simulation (DNS) databases provide proper initial and boundary conditions for the NSFnet simulations. The spatial and temporal coordinates are the inputs of the NSFnets, while the instantaneous velocity and pressure fields are the outputs for the VP-NSFnet, and the instantaneous velocity and vorticity...

[144] [Error estimates of residual minimization using neural networks for linear PDEs](#)

Yeonjong Shin, ..., and G. Karniadakis. ArXiv, 2020. 93 citations.

Not measured Topic Match

Abstract: We propose an abstract framework for analyzing the convergence of least-squares methods based on residual minimization when feasible solutions are neural networks. With the norm relations and compactness arguments, we derive error estimates for both continuous and discrete formulations of residual minimization in strong and weak forms. The formulations cover recently developed physics-informed neural networks based on strong and variational formulations.

[145] [Solving PDEs by variational physics-informed neural networks: an a posteriori error analysis](#)

S. Berrone, ..., and Moreno Pintore. ANNALI DELL'UNIVERSITA' DI FERRARA, 2022. 31 citations.

Not measured Topic Match

Abstract: We consider the discretization of elliptic boundary-value problems by variational physics-informed neural networks (VPINNs), in which test functions are continuous, piecewise linear functions on a triangulation of the domain. We define an a posteriori error estimator, made of a residual-type term, a loss-function term, and data oscillation terms. We prove that the estimator is both reliable and efficient in controlling the energy norm of the error between the exact and VPINN solutions. Numerical results are in excellent agreement with the theoretical predictions.

[146] [Characterizing possible failure modes in physics-informed neural networks](#)

Aditi S. Krishnapriyan, ..., and Michael W. Mahoney. Unknown journal, 2021. 1191 citations.

Not measured Topic Match

Abstract: Recent work in scientific machine learning has developed so-called physics-informed neural network (PINN) models. The typical approach is to incorporate physical domain knowledge as soft constraints on an empirical loss function and use existing machine learning methodologies to train the model. We demonstrate that, while existing PINN methodologies can learn good models for relatively trivial problems, they can easily fail to learn relevant physical phenomena for even slightly more complex problems. In particular, we analyze several distinct situations of widespread physical interest, including learning differential equations with convection, reaction, and diffusion operators. We provide evidence that the soft regularization in...

[147] [Variationally Correct Neural Residual Regression for Parametric PDEs: On the Viability of Controlled Accuracy](#)

M. Bachmayr, ..., and Mathias Oster. ArXiv, 2024. 13 citations.

Not measured Topic Match

Abstract: This paper is about learning the parameter-to-solution map for systems of partial differential equations (PDEs) that depend on a potentially large number of parameters covering all PDE types for which a stable variational formulation (SVF) can be found. A central constituent is the notion of variationally correct residual loss function meaning that its value is always uniformly proportional to the squared solution error in the norm determined by the SVF, hence facilitating rigorous a posteriori accuracy control. It is based on a single variational problem, associated with the family of parameter dependent fiber problems, employing the notion of direct integrals...

[148] [The Deep Ritz Method: A Deep Learning-Based Numerical Algorithm for Solving Variational Problems](#)

E. Weinan and Ting Yu. Communications in Mathematics and Statistics, 2017. 1835 citations.

Not measured Topic Match

Abstract: We propose a deep learning-based method, the Deep Ritz Method, for numerically solving variational problems, particularly the ones that arise from partial differential equations. The Deep Ritz Method is naturally nonlinear, naturally adaptive and has the potential to work in rather high dimensions. The framework is quite simple and fits well with the stochastic gradient descent method used in deep learning. We illustrate the method on several problems including some eigenvalue problems.

[149] [DPM: A Novel Training Method for Physics-Informed Neural Networks in Extrapolation](#)

Jungeun Kim, ..., and Noseong Park. ArXiv, 2020. 114 citations.

Not measured Topic Match

Abstract: We present a method for learning dynamics of complex physical processes described by time-dependent nonlinear partial differential equations (PDEs). Our particular interest lies in extrapolating solutions in time beyond the range of temporal domain used in training. Our choice for a baseline method is physics-informed neural network (PINN) because the method parameterizes not only the solutions, but also the equations that describe the dynamics of physical processes. We demonstrate that PINN performs poorly on extrapolation tasks in many benchmark problems. To address this, we propose a novel method for better training PINN and demonstrate that our newly enhanced PINNs can...

[150] [An Introductory Review on A Posteriori Error Estimation in Finite Element Computations](#)

Ludovic Chamoin and F. Legoll. SIAM Rev., 2023. 20 citations.

Not measured Topic Match

No summary or abstract available

[151] [A note on Strichartz estimates for the wave equation with orthonormal initial data \(Harmonic Analysis and Nonlinear Partial Differential Equations\)](#)

N. Bez, ..., and Shobu Shiraki. RIMS Kôkyûroku Bessatsu, 2023. 4 citations.

Not measured Topic Match

Abstract: This note is concerned with Strichartz estimates for the wave equation and orthonormal families of initial data. We provide a survey of the known results and present what seems to be a reasonable conjecture regarding the cases which have been left open. We also provide some new results in the maximal-in-space boundary cases.

[152] [Surrogate modeling for fluid flows based on physics-constrained deep learning without simulation data](#)

Luning Sun, ..., and Jian-Xun Wang. Computer Methods in Applied Mechanics and Engineering, 2019. 1024 citations.

Not measured Topic Match

Abstract: Abstract Numerical simulations on fluid dynamics problems primarily rely on spatially or/and temporally discretization of the governing equation using polynomials into a finite-dimensional algebraic system. Due to the multi-scale nature of the physics and sensitivity from meshing a complicated geometry, such process can be computational prohibitive for most real-time applications (e.g., clinical diagnosis and surgery planning) and many-query analyses (e.g., optimization design and uncertainty quantification). Therefore, developing a cost-effective surrogate model is of great practical significance. Deep learning (DL) has shown new promises for surrogate modeling due to its capability of handling strong nonlinearity and high dimensionality. However, the off-the-shelf...

[153] [Certified Reduced Basis Methods for Parametrized Partial Differential Equations](#)

J. Hesthaven, ..., and B. Stamm. Unknown journal, 2015. 1130 citations.

Not measured Topic Match

No summary or abstract available

[154] [Building Trust in PINNs: Error Estimation through Finite Difference Methods](#)

Aleksander Krasowski, ..., and J. Naujoks. ArXiv, 2026. 0 citations.

Not measured Topic Match

Abstract: Physics-informed neural networks (PINNs) constitute a flexible deep learning approach for solving partial differential equations (PDEs), which model phenomena ranging from heat conduction to quantum mechanical systems. Despite their flexibility, PINNs offer limited insight into how their predictions deviate from the true solution, hindering trust in their prediction quality. We propose a lightweight post-hoc method that addresses this gap by producing pointwise error estimates for PINN predictions, which offer a natural form of explanation for such models, identifying not just whether a prediction is wrong, but where and by how much. For linear partial differential equations, the error between a...

[155] [Man, Machine, and Mathematics](#)

Akshunna S. Dogra. ArXiv, 2026. 0 citations.

Not measured Topic Match

Abstract: Nonlinear models and optimization methods have successfully tackled a rapidly growing set of problems in recent years. Indeed, a relatively small toolbox of such models and methods can provide sufficient performance across a large landscape of tasks: deep learning alone has made significant recent contributions in scientific modelling, natural language processing, visual analysis, etc. A similar relationship exists between physical theories and phenomena, where many applications and observations emerge neatly from remarkably minimal foundations. It is natural to wonder if sparse unified frameworks could be built to steer discussion and discovery in the fields concerned with learning, optimization, and modelling...

[156] [Estimation of the input-to-state stability gain functions from finite-dimensional approximations](#)

Birgit Hillebrecht and Benjamin Unger. ArXiv, 2026. 0 citations.

Not measured Topic Match

Abstract: Since the concept of input-to-state stability (ISS) was introduced, it has been extensively investigated for finite-dimensional control systems and has recently received attention for infinite-dimensional systems. While numerical techniques provide a bridge between these two worlds, a rigorous connection between the ISS of an infinite-dimensional system with an unbounded control operator and the properties of its finite-dimensional approximations has not yet been established. In this manuscript, we make a first step towards closing this gap by investigating numerical approximations of linear (boundary) control systems using semigroup theory. Specifically, we focus on linear boundary control systems where the autonomous evolution is...

[157] [Fredholm Neural Networks for inverse problems in elliptic PDEs](#)

Kyriakos Georgiou, ..., and A. Yannacopoulos. ArXiv, 2025. 1 citations.

Not measured Topic Match

Abstract: Building on our previous work on Fredholm Neural Networks (Fredholm NNs/ FNNs) for solving integral equations, we extend the framework to inverse problems for linear and nonlinear elliptic partial differential equations. The proposed scheme consists of a custom-designed deep neural network (DNN) in which

the number of layers, weights, biases and hyperparameters are computed in an explainable manner based on a fixed-point scheme, and we therefore refer to this as the Potential Fredholm Neural Network (PFNN). We first build the PFNN as a method for solving the forward problem, showing that this approach ensures both a high accuracy and explainability,...

[158] [Physics-Informed Neural Networks for the Structural Analysis and Monitoring of Railway Bridges: A Systematic Review](#)

Yuniel Martínez, ..., and José García. Mathematics, 2025. 23 citations.

Not measured Topic Match

Abstract: Physics-informed neural networks (PINNs) offer a mesh-free approach to solving partial differential equations (PDEs) with embedded physical constraints. Although PINNs have gained traction in various engineering fields, their adoption for railway bridge analysis remains under-explored. To address this gap, a systematic review was conducted across Scopus and Web of Science (2020–2025), filtering records by relevance, journal impact, and language. From an initial pool, 120 articles were selected and categorised into nine thematic clusters that encompass computational frameworks, hybrid integration with conventional solvers, and domain decomposition strategies. Through natural language processing (NLP) and trend mapping, this review evidences a growing but...

[159] [Advancing Generalization in PINNs Through Latent-Space Representations](#)

Honghui Wang, ..., and Gao Huang. IEEE Transactions on Neural Networks and Learning Systems, 2024. 7 citations.

Not measured Topic Match

Abstract: Physics-informed neural networks (PINNs) have made significant strides in modeling dynamical systems governed by partial differential equations (PDEs). However, their generalization capabilities across varying scenarios remain limited. To overcome this limitation, we propose physics-informed dynamics representation learner (PiDo), a novel physics-informed neural PDE solver designed to generalize effectively across diverse PDE configurations, including varying initial conditions, PDE coefficients, and training-time horizons. PiDo exploits the shared underlying structure of dynamical systems with different properties by projecting PDE solutions into a latent space using auto-decoding. It then learns the dynamics of these latent representations, conditioned on the PDE coefficients. Despite its promise,...

[160] [Magnetic Field Analysis of Reactor Based on TL-PIDL and Time Domain PIDL](#)

Guang Yang, ..., and Jiahao Liu. IEEE Transactions on Applied Superconductivity, 2024. 1 citations.

Not measured Topic Match

Abstract: With the widespread use of deep learning in various fields, physics informed deep learning (PIDL) is proposed to be applied in a magnetic field problem. To meet the needs of practical engineering, we predict the electromagnetic response of the reactor by combining the theory related to magnetic fields with PIDL, and its feasibility is verified by comparing its predictions to those of finite element methods (FEM). Subsequently, by using transfer learning (TL), it is possible to perform quick analysis based on analytical experience before changing the geometry. Lastly, the performance of the proposed PIDL was validated using results obtained from...

[161] [PINNs error estimates for nonlinear equations in -smooth Banach spaces](#)

Jiexing Gao and Yurii Zakharian. ArXiv, 2023. 2 citations.

Not measured Topic Match

Abstract: In the paper, we describe in operator form classes of PDEs that admit PINN's error estimation. Also, for  $L^p$  spaces, we obtain a Bramble-Hilbert type lemma that is a tool for PINN's residuals bounding.

[162] [Port-Hamiltonian Dynamic Mode Decomposition](#)

R. Morandin, ..., and B. Unger. SIAM J. Sci. Comput., 2022. 21 citations.

Not measured Topic Match

Abstract: We present a novel physics-informed system identification method to construct a passive linear time-invariant system. In more detail, for a given quadratic energy functional, measurements of the input, state, and output of a system in the time domain, we find a realization that approximates the data well while guaranteeing that the energy functional satisfies a dissipation inequality. To this end, we use the framework of port-Hamiltonian (pH) systems and modify the dynamic mode decomposition, respectively operator inference, to be feasible for continuous-time pH systems. We propose an iterative numerical method to solve the corresponding least-squares minimization problem. We construct an...

[163] [Physics-informed deep learning for elasto-plasticity: Strong-form, energy-minimization, and state-aware formulations](#)

Sumit Kumar and I. Singh. Finite Elements in Analysis and Design, 2026. 0 citations.

Not measured Topic Match

No summary or abstract available

[164] [Learn and Verify: A Framework for Rigorous Verification of Physics-Informed Neural Networks](#)

Kazuaki Tanaka and Kohei Yatabe. ArXiv, 2026. 1 citations.

Not measured Topic Match

Abstract: The numerical solution of differential equations using neural networks has become a central topic in scientific computing, with Physics-Informed Neural Networks (PINNs) emerging as a powerful paradigm for both forward and inverse problems. However, unlike classical numerical methods that offer established convergence guarantees, neural network-based approximations typically lack rigorous error bounds. Furthermore, the non-deterministic nature of their optimization makes it difficult to mathematically certify their accuracy. To address these challenges, we propose a "Learn and Verify" framework that provides computable, mathematically rigorous error bounds for the solutions of differential equations. By combining a novel Doubly Smoothed Maximum (DSM) loss for training with...

[165] [Multi-Scale Finite Expression Method for PDEs with Oscillatory Solutions on Complex Domains](#)

Gareth Hardwick and Haizhao Yang. ArXiv, 2025. 0 citations.

Not measured Topic Match

Abstract: Solving partial differential equations (PDEs) with highly oscillatory solutions on complex domains remains a challenging and important problem. High-frequency oscillations and intricate geometries often result in prohibitively expensive representations for traditional numerical methods and lead to difficult optimization landscapes for machine learning-based approaches. In this work, we introduce an enhanced Finite Expression Method (FEX) designed to address these challenges with improved accuracy, interpretability, and computational efficiency. The proposed framework incorporates three key innovations: a symbolic spectral composition module that enables FEX to learn and represent multiscale oscillatory behavior; a redesigned linear input layer that significantly expands the expressivity of the...

[166] [Evolution of linear matter perturbations with error-bounded bundle physics-informed neural networks](#)

Luca Gomez Bachar, ..., and P. Protopapas. Physical Review D, 2025. 1 citations.

Not measured Topic Match

Abstract: We consider the evolution of linear matter perturbations in the context of the standard cosmological model ( $\Lambda$ CDM) and a phenomenological modified gravity model. We use the physics-informed neural network (PINN) bundle method, which allows to integrate differential systems as an alternative to the traditional numerical method. We apply the PINN bundle method to the equation that describes the matter perturbation evolution, to compare its outcomes with recent data on structure growth,  $\sigma_8$ . Unlike our previous works, we can calculate a bound on the error of this observable without using the numerical solution of the equation. For this, we use a...

[167] [Improved Uncertainty Quantification in Physics-Informed Neural Networks Using Error Bounds and Solution Bundles](#)

P. Flores, ..., and K. Pichara. ArXiv, 2025. 4 citations.

Not measured Topic Match

Abstract: Physics-Informed Neural Networks (PINNs) have been widely used to obtain solutions to various physical phenomena modeled as Differential Equations. As PINNs are not naturally equipped with mechanisms for Uncertainty Quantification, some work has been done to quantify the different uncertainties that arise when dealing with PINNs. In this paper, we use a two-step procedure to train Bayesian Neural Networks that provide uncertainties over the solutions to differential equation systems provided by PINNs. We use available error bounds over PINNs to formulate a heteroscedastic variance that improves the uncertainty estimation. Furthermore, we solve forward problems and utilize the obtained uncertainties when...

[168] [Augmented neural forms with parametric condition-matching operators for solving ordinary differential equations](#)

Adam D. Kyriadiis, ..., and K. Parsopoulos. Machine Learning for Computational Science and Engineering, 2024. 0 citations.

Not measured Topic Match

No summary or abstract available

- [169] [Augmented neural forms with parametric boundary-matching operators for solving ordinary differential equations](#)

Adam D. Kypriadis, ..., and K. Parsopoulos. ArXiv, 2024. 0 citations.

Not measured Topic Match

Abstract: Approximating solutions of ordinary and partial differential equations constitutes a significant challenge. Based on functional expressions that inherently depend on neural networks, neural forms are specifically designed to precisely satisfy the prescribed initial or boundary conditions of the problem, while providing the approximate solutions in closed form. Departing from the important class of ordinary differential equations, the present work aims to refine and validate the neural forms methodology, paving the ground for further developments in more challenging fields. The main contributions are as follows. First, it introduces a formalism for systematically crafting proper neural forms with adaptable boundary matches that...

- [170] [One-Shot Transfer Learning for Nonlinear ODEs](#)

Wanzhou Lei, ..., and Joy Parikh. ArXiv, 2023. 3 citations.

Not measured Topic Match

Abstract: We introduce a generalizable approach that combines perturbation method and one-shot transfer learning to solve nonlinear ODEs with a single polynomial term, using Physics-Informed Neural Networks (PINNs). Our method transforms non-linear ODEs into linear ODE systems, trains a PINN across varied conditions, and offers a closed-form solution for new instances within the same non-linear ODE class. We demonstrate the effectiveness of this approach on the Duffing equation and suggest its applicability to similarly structured PDEs and ODE systems.

- [171] [Two-Step Bayesian PINNs for Uncertainty Estimation](#)

P. Flores, ..., and K. Pichara. Journal Not Provided, Unknown year. 0 citations.

Not measured Topic Match

No summary or abstract available

- [172] [Numerical Study of Neural Network Surrogate Model Performance for Uncertainty Propagation](#)

Noah Wade and K. Teferra. Journal of Engineering Mechanics, 2026. 0 citations.

Not measured Topic Match

No summary or abstract available